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RESEARCH ARTICLE

Determinants of Supply Response of Coconut Cultivation in Sri Lanka

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ABSTRACT

Coconut (*Cocos nucifera* L) is a major plantation crop grown in Sri Lanka. The output has been mainly used for local consumption of fresh nuts, coconut-based industries within the country, and coconut exports. The instability of coconut supply seriously affects local as well as export market potentials. Identification of various factors and the extent to which those affected the current coconut supply have not been examined in the recent empirical work. This study examines the supply response of coconut in Sri Lanka, using Nerlovian Partial Adjustment model. The study used time series data of total coconut production, output price, rainfall, fertilizer amount, land extent, technology, and the presence of fertilizer subsidy for a period of 30 years (1985-2015) for the analysis. Among the different types of functional forms tested, the log-linear model was selected as the best fit model ($R^2=0.894$). The model was tested for autocorrelation and heteroscedasticity using the Durbin-Watson statistic and the Breusch-pagan test, respectively. The results revealed that the variables, i.e. two years lagged price, one year lagged rainfall, and two years lagged fertilizer amount affect the coconut production significantly ($p<0.05$). The short-run and long-run price elasticities were estimated for two years lagged output price. The output price is the most influential factor in short run with the elasticity of 0.33 (two year lagged price variables were considered) while rainfall is the most important determinant of coconut production in Sri Lanka. The study concludes that coconut production in the country tends to change as a response to the changes in coconut price both in the short-run and long-run. Further, rainfall and amount of fertilizer applied are the two main factors that determine the coconut supply in Sri Lanka.

Keywords: *Cocos nucifera*, Nerlovian partial adjustment model, Supply response

INTRODUCTION

The coconut tree (*Cocos nucifera* L) is a member of the family Arecaceae (palm family) and the only species of the genus *Cocos*. Coconut seed is the major item of coconut and botanically it is a drupe that gives multiple uses. Coconut palms are grown in more than 90 countries of the world, with an average total fresh nut production of 61 million tonnes per year. Sri Lanka is the fourth largest coconut producer in the world and it contributes 4.3% of total world coconut production (FAOSTAT, 2020).

Coconut is the largest plantation crop in Sri Lanka and the annual coconut production in 2022 was 335 million nuts. In 2022, contribution of coconut cultivation to the Gross Domestic Product (GDP) was 0.9%, while agricultural contribution to the GDP was 7.5% (Central Bank, 2022). Coconut plays an important role in developing the socio-economic condition of the people as it has multiple uses. It is an essential component of the Sri Lankan cuisine, nutrition, and rural livelihood as it is a major component in the Sri Lankan diet. An average household spends about 5.4 % of the total expenditure on food and drinks on coconut, which is the principal source of edible oil and fats (Department of Census and Statistics, 2019). The per capita coconut consumption including coconut oil is 122 nuts per year (Department of Census and Statistics, 2019). In addition, coconut-based industries contribute substantially to employment generation and foreign exchange earnings. Domestic fresh coconut consumption absorbs nearly 56% of the total production in the last five years and the rest is used for processed coconut products (Coconut Development Authority, 2021).

Annual coconut production in the past few years revealed that coconut production in Sri Lanka is in slow growth and it is a serious impediment to the economic viability of the coconut industry. The average coconut yields have been stagnant over time but there is a substantial year-to-year variability due to climatic and other factors as application of fertilizer and mulching. This instability of coconut supply is a crucial issue in the coconut industry. These factors include product's own price, rainfall, fertilizer application, presence of fertilizer subsidy, and cultivated land extent. The short supply of coconut creates many consequences in the society as coconut is one of the main items in the Sri Lankan food basket, economy and livelihood. Reduction of coconut supply leads to increased coconut prices. Low production and high coconut prices badly affect the day to day consumption, coconut-based industries, and finally, it causes to reduce export income earnings from the coconut industry. On the other hand, it affects future production as producers also respond to the changes in coconut price.

Supply response explains the change in output due to changes in output price and other factors. It helps one to ascertain how farmers reallocate resources in response to the changes in relative price levels. Information on supply price elasticity allows the formulation of appropriate agricultural price policies and helps to understand and predict the short-run and long-run impact of price changes on production. When there is a crisis with production and supply of coconut, the government and the responsible parties take several short-term remedies but attention towards long-term measures in order to increase production is a must where

factors affecting supply response plays an important role. However, recent literature on supply response of coconut with the price and other factors are lacking. Hence, this research on coconut supply response is a timely vital need.

The general objective of this study is to examine the supply response of coconut production in Sri Lanka. Specifically, the study estimates both short-run and long-run price elasticities of fresh coconut production. Furthermore, it estimates the extent to which the factors namely fertilizer used, cultivated land area, rainfall, and presence of fertilizer subsidy affect the coconut production of Sri Lanka.

The remainder of the paper is structured as follows. Section II reviews literature on the supply response of perennial crops. Materials and methods used in the study are presented in the section III. Section IV explains and discusses the results of the empirical analysis while section V presents conclusions of the present study.

2. Literature Review

Supply response is the change in the quantity of a commodity output forward for sale when other things are not held constant and it is considered with the output response to a price change (Cochrane, 1955). Estimation of supply response in perennial crops needs special consideration as output flows over many years and in coconuts, it is about 60 years. The present output of a perennial crop is determined by the planting decisions in the past. Growers can change their production either by increasing land extent under cultivation or re-planting the senile palms. In perennial crops, there is a lag between inputs

and outputs. In addition, technology improvement and the adoption of improved inputs increase production capacity. Hence, the present production capacity is the net effect of past decisions that change the production capacity of the plantation (Kalaitzandonakes & Shonkwiler, 1992). Therefore, estimation of supply response of perennials must include the new plantings, removals, replanting, age composition, and the lag between inputs and output. However, the dearth of long-term time-series data on most of these factors compelled most of the studies on perennial supply response to use aggregate data.

Perennial crops have two distinctive features; i.e.(1) gestation period between planting and the first harvest which is determined by the biological nature of the crop and (2) significant adjustment cost of planting and removal which is determined by the decisions in the past, technology, availability of land, labour and capital (French & Matthews, 1971; Akiyama & Trivedi, 1987). The gestation period of coconut vary around 2-5 years depending on the variety.

The estimation of supply response to prices was advanced after the seminal work of Mark Nerlove (1958). This Nerlove model was primarily used for annual crops but later, it was modified to analyse perennial crops. The first attempt to adopt the dynamic Nerlove framework to study the supply response of perennial was employed by Bateman (1965) for cocoa in Ghana. The area is considered a function of discounted expected own and substitute prices. Similar Nerlovian models were used by Ady (1968) and Behrman (1968) for cocoa. The difference is that the desired acreage is modeled as a function of expected own and substitute prices instead of actual

acreage. French and Mathews (1971) in their seminal paper on perennial supply response, developed a theory to explain new plantings and acreage adjustment. This is empirically applied to the asparagus industry in the USA and specified acreage was taken as a function of lagged own price and lagged acreage. The above studies laid a theoretical foundation for the analysis of supply response for perennials. This model was extended later with modifications.

Samarajeewa *et al.*, (2001) estimated the supply response of coconut production in Sri Lanka. The Nerlovian partial adjustment model was used to estimate the regression coefficient of a specific equation. The model was corrected for autocorrelation and heteroscedasticity to obtain consistent results. The results revealed that the coconut producers respond to output price changes significantly in the short-run than in the long-run. Two-year lagged price, one-year lag rainfall, and cost of production of the current year are the crucial factors that affect coconut production.

Leaver (2004) evaluated the supply response of Tobacco in Zimbabwe. The analysis was carried out using the data from 1938 to 2000. A modified form of the Nerlovian model was used in this study to prevent various problems that arise from the statistical estimation of the original Nerlovian model. The results show that tobacco farmers are highly unresponsive to price changes.

Karunakaran & Gangadharan (2014) estimated the supply response of coconut in Kerala. Nerlovian supply response model was used in this study and the farmers' decisions are discussed in terms of area response and yield. The analysis revealed

that the irrigated area, rainfall, and price risk factors are the significant variables affecting the area allocation for coconut in Kerala and the crop's yield response, irrigated area, rainfall and expected price risk are the most significant variables.

3. Materials and Methods

3.1. Data

According to the objectives of the study, the supply response of coconut was analyzed for a period of 31 years from 1985 to 2015. Secondary data were obtained from the annual reports of Central Bank, Coconut Statistics of the Coconut Development Authority and General Reports of Census of Agriculture. Use of the nominal output price does not make economic sense if inflation is high, since farmers will be interested in the actual purchasing power of their money and as a result will respond to changes in real output prices rather than changes in nominal prices (Leaver, 2004). Accordingly, as this study uses the time series data of price, they need to be adjusted for inflation. Hence nominal prices of coconut were converted into real prices using the Wholesale Price Index (WPI) of each year using the following formula.

$$Real\ price_z = Nominal\ price_z \times Adjustment\ factor \quad (1)$$

The adjustment factor is formed using WPI measures as follows.

$$Real\ price_z = Nominal\ price_z \times \frac{WPI_{Base\ year}}{WPI_z} \quad (2)$$

Where,

Z = year considered

3.2. The Model

Supply response analyses were conducted in two frameworks: (1) the Nerlovian model which facilitates a level of adjustment and speed of adjustment of actual acreage and desired acreage and (2) supply function derived from profit-maximizing framework (Dasgupta, 2009; Vrankic & Krpan, 2017). Joint estimation of output supply and input demand is used in the latter method. This type of analysis requires information on all input prices. In addition, market intervention for the input market is very common in Sri Lanka and the input market for agricultural products are not competitive. These factors led to the selection of the Nerlovian partial adjustment model for this analysis and supply response was measured in terms of quantity response to prices.

According to Nerlove, the desired output can be expressed as a function of the expected price and exogenous shifters (Cummings & Askari, 1977).

$$Q_t^* = a + bP_t^* + cZ_t \quad (3)$$

where, Q_t^* is desired output in period t , P_t^* expected price in period t , and Z_t is a set of exogenous shifters in period t such as technology change, weather condition etc.

Actual output may differ from the desired because of the adjustment lags of variable factors. Therefore, it is assumed that the actual output would only be a fraction d of the desired output.

$$Q_t - Q_{t-1} = d(Q_t^* - Q_{t-1}) \quad (4)$$

where, Q_t is the actual output in period t , Q_{t-1} is the actual output in period $t-1$, and d is the adjustment coefficient of output. Its value lies between 0 and 1.

The farmers' expected price at harvest time can be observed. So, it needs to describe how the decision-makers form expectations based on the knowledge of actual and past prices and other observable information. It is assumed that farmers remember in their memory the magnitude of the mistakes they made in the previous period and learned by adjusting the difference between the actual and expected price in $t-1$ by a fraction λ .

$$P_t^* = \lambda P_{t-1} + (1-\lambda) P_t^* - 1 \quad (5)$$

The adjustment parameter λ is the expectation coefficient of price. It measures the speed of adjustment of the expected price and actual price and the value lies between 0 and 1. The value closer to 1 indicates the faster the speed of adjustment.

In order to estimate the supply response using the Nerlovian model, it is necessary to eliminate the unobservable variables associated with expected price and the desired output from equations (3) to (5) (Braulke, 1982). By eliminating these variables, the estimating or "reduced form" Nerlovian equation is achieved.

Using the values of P_t^* and Q_t^* from equation 4 and 5, the equation 3 can be written as;

$$Q_t = ad\lambda + bd\lambda P_{t-1} + (1-d) + (1-\lambda) Q_{t-1} + cdZ_t \quad (6)$$

$$Q_t = A_0 + A_1 P_{t-1} + A_2 Q_{t-1} + A_3 Z_t \quad (7)$$

where, A_0 is $ad\lambda$, A_1 is $bd\lambda$, A_2 is $(1-d) + (1-\lambda)$, and A_3 is cd .

The biological nature of the coconut crop identifies different time lags between dependent and independent variables. The study uses time-series data to predict the current values of the dependent variable (coconut supply) based on both the current values and the lagged (past period) values of

explanatory variables. Hence, the distributed lag model was used for the analysis. Dependent variable Y was determined by the weighted sum of the past values (lagged) of independent variables, Xs (Almon, 1965). It defined the pattern of how Xs affect Y over time.

According to this model, the short-run and long-run price elasticities of supply are computed simultaneously. Elasticity formulas for the linear model can be expressed as follows. However, this changes according to the functional form that is used.

$$\text{Short - run Price Elasticity} = A_1 \times \frac{P_{t-1} \text{ mean}}{Q_{t-1} \text{ mean}} \quad (8)$$

$$\text{Long - run Price Elasticity} = \frac{A_1}{1-A_2} \times \frac{P_{t-1} \text{ mean}}{Q_{t-1} \text{ mean}} \quad (9)$$

$$\text{Short - run Price Elasticity} = A_1 \times \text{Pt-1 mean} / \text{Qt-1 mean}$$

The empirical model selected for the study is represented by the following function in the linear form but the different models will be estimated and the best fit model is selected.

$$Q_t = a + bQ_{(t-p)} + bP_{(t-q)} + cFA_{(t-r)} + dRF_{(t-s)} + eL_t + gFS_t + hT + U(10)$$

Where,

Q_t = Production (nuts in million)

P = Real price of coconut (Rs/1000 nuts)

FA = Fertilizer amount (tonnes)

RF = Rainfall (mm)

L = Land extent (ha)

FS = Presence of fertilizer subsidy (Dummy variable; given=1 or zero otherwise)

T = Time (years)

U = Error term

$(t-p), (t-q), (t-r), (t-s)$ = Time lags

a, b, c, d, e, g, h = Coefficients of variables

It is assumed that the desired level of output in time t is a function of lagged output and lagged forms of output price, rainfall, land extent, presence of fertilizer subsidy, time, and the fertilizer amount.

The total coconut production was regressed with independent variables in the model to uncover the best fit variables. Different types of functional forms such as linear, log-linear, lin-log, and log-log were tested. The fitted model was estimated by the Ordinary Least Square (OLS) test using the Stata version 14.2 statistical software. An autocorrelation test was conducted using the Durbin-Watson statistic to test the autocorrelation effect. At the same time, the Bresch-pagan/Cook-Weisberg test was conducted to test for heteroskedasticity.

4. Results and Discussion

The suitability of the models were judged based on Akaike's Information Criterion (AIC), coefficient of determination (R^2), expected signs of the estimated parameters, and by the significance of the independent variables. Accordingly, the Log-linear model was selected as the best model.

The regression results of the Log-linear model are presented in Table 1. Given test statistics for overall significance of the model follow F-distribution and it appeared that the estimations are highly significant. The R^2 value indicates that approximately 89% of the variation is explained by the model. Except for one year lagged and three year lagged fertilizer amount applied, all the other variables bear expected signs and hence they are in the right direction and in lined with the

theoretical justifications. The results show that two year lagged price, one year lagged rainfall, and two year lagged fertilizer amount are statistically significant at a 5% significant level and show a positive relationship with coconut production.

Table 1: Coefficient estimates of Log-linear model

Variables	Coefficient estimates	<i>t</i> Statistic	<i>P</i> Value
$Q_{(t-1)}$	0.164e-04	1.22	0.828
$P_{(t-2)}$	0.303e-04	2.36	0.035*
$RF_{(t-1)}$	0.122e-03	2.91	0.012*
$FA_{(t-2)}$	0.532e-07	2.59	0.022*
L_t	0.37e09	0.08	0.941
FS_t	0.508376	1.91	0.078
T	0.001861	0.54	0.598
<i>Cons</i>	6.585567	22.74	0.000

*indicates significant at 5% level.

Table 2: Short-run and Long-run Elasticities

Variables	Short-run Elasticity	Long-run Elasticity
$P_{(t-2)}$	0.33	0.08
$RF_{(t-1)}$	0.23	0.32
$FA_{(t-2)}$	0.18	0.01

Short-run elasticity of two year lagged coconut price was 0.33 (Table 2). It implies that when there is a rise in the current coconut price by one percent, the coconut production may increase by 0.33% in the year after next year. The short run and long run elasticities of one-year lagged rainfall was reported as 0.23 and 0.01 respectively. It indicates that if there is an increase in rainfall by one percent in the current year, coconut production will increase by 0.23% year after. The elasticity of two year lagged fertilizer amount is 0.18 indicating one percent increased fertilizer application in the current year results in a 0.18% rise in coconut

production after two years. As supply elasticities of coconut price, rainfall, and fertilizer amount are less than one, it can be concluded that they are inelastic.

Results indicate that the coconut supply of Sri Lanka is responsive to price changes. As there is no any fixed level of coconut price in Sri Lanka, it has been fluctuating in a broad range for many years. When supply of a commodity is decreased, its price is increased and this is applied to coconuts as well. Although the coconut price are increased, consumers tend to purchase as it is a major component in Sri Lankan food items.

Rainfall is one of the most influential factors on coconut supply of Sri Lanka. A well-distributed mean annual rainfall of between 1,500-2,000 mm is considered to be favorable for the cultivation of coconut. According to the summary statistics, within the past 31 years, rainfall has fluctuated between a range of 1,363.33-2,288.5 mm. Hence, coconut supply also fluctuated accordingly. The coconut bunch goes through a cycle of development lasting about 36 months before it is ready to harvest. The development of nuts takes place during the last twelve months after the opening of the inflorescence (Child, 1974). During the nut developing period, especially three months after flower opening of the coconut as well as three months before flower opening are very sensitive to major climatic elements namely rainfall and temperature. Hence, the coconut supply depends on the change of rainfall significantly. Effect of current weather is usually reflected on the yields of the following year due to its lagged effects.

Use of fertilizer is one of the effective short term method of increasing coconut production in Sri Lanka and it is one of the main factors which affects yield in the short run under the control of farmer (Silva, 1981). The removal of nuts from estates removes a considerable amount of nutrition from coconut plantations. In addition to the removal by palms, soil nutrients can be lost by leaching, runoff,

erosion, and uptake by weeds. The above processes lead to the depletion of essential plant nutrients in the soil, which causes a gradual decline of nut yield. Therefore, soils in coconut plantations should be enriched with nitrogen (urea), phosphorus (Eppawala rock phosphate), potassium (muriate of potash), and magnesium (dolomite) which are the major nutrients that remove from above mentioned processes (Mahindapala and Pinto,1986). The nutrient requirement of young coconut palms and adult palms are different and therefore different fertilizer recommendations are made to them separately. Thus, coconut farmers tend to change the coconut supply by changing the fertilizer application whenever necessary. Summary statistics show a greater variation of fertilizer application with the minimum amount of 5,456 tons to maximum amount of 47,114 tons within the past 31 years by resulting in a supply response of coconut to change of fertilizer application. Due to the perennial nature of coconut, and the rate of release of nutrients from fertilizer, it takes two years to increase the coconut supply as a response to the increase in fertilizer application.

Data were tested for heteroskedasticity and autocorrelation using Breuch-pagan test and Durbin-Watson statistics, respectively. The results of these tests are presented in Table 3 and 4, respectively.

Table 3: Results of Heteroskedasticity test

Breusch-pagan / Cook-Weisberg test for heteroskedasticity		
H0 : Constant variance		
Variables : fitted values of coconut production (nuts/mn)		
Chi2(1)	=	0.05
Prob > chi2	=	0.8159

The null hypothesis is not rejected, as the probability value is greater than the conventional trigger point ($\alpha = 0.05$) indicating that the variance is constant and data are free from heteroskedasticity.

Table 4: Results of Durbin-Watson statistics

Number of gaps in sample	:	2
Durbin-Watson d-statistic (14, 20)	=	2.118134

A rule of thumb is that test statistic values in the range of 1.5 to 2.5 are relatively normal. As the Durbin-Watson statistic is close to two, there is no autocorrelation. The error terms of data do not show a significant correlation with each other.

5. Conclusions and Suggestions

This study investigated the supply response of coconut to price fluctuations and changes in rainfall, amount of fertilizer applied, presence of fertilizer subsidy, technology, and the land extent. The findings of this study revealed that coconut production responds significantly to output price fluctuations and it is the most influential factor on coconut production in Sri Lanka. The coconut supply of Sri Lanka positively responds to two years lagged coconut prices. Among the other factors that significantly affect supply response of coconut output in Sri Lanka are one-year lagged rainfall, and two-year lagged fertilizer amount. All these factors are positively affecting coconut production and the higher the above factors, the greater the supply of coconut in Sri Lanka.

The elasticity of two-year lagged coconut price is 0.33 in the short run while it is 0.08 in the long run. The elasticity of one-year lagged rainfall is 0.23 in the short run while it is 0.32 in the long run. The elasticity of the two-year lagged fertilizer amount is 0.18 in the short run while it is 0.01 in the long run.

The findings can be used by the government and policymakers for price setting for coconut and to implement agricultural and income policies to encourage greater production of coconut in the country.

REFERENCES

- Akiyama, T. & Trivedi, P.K. (1987). Vintage production approach to perennial crop supply: An application to tea in major producing countries. *Journal of Econometrics*, 36(1-2), 133-161.
- Almon, S. (1965). The Distributed Lag Between Capital Appropriations and Expenditures. *Econometrica*, 33(1), 178-196.
- Ady, P. (1968). Supply functions in tropical agriculture *Oxford Bulletin of Economics and Statistics*, 30(2), 157-188.
- Bateman, M. J. (1965). Aggregate and regional supply functions for Ghanaian cocoa. 1946-1962, *Journal of Farm Economics*, 47(2), 384-401.
- Behrman, J. R. (1968). Monopolistic cocoa pricing *American Journal of Agricultural Economics*, 50(3), 702-719.

- Braulke, M. (1982). A note on the Nerlove model of agricultural supply response. *International Economic Review*, 23 (1), 241-244.
- Central Bank of Sri Lanka. (2018). External Sector Developments and Policies. Central Bank of Sri Lanka. Retrieved May 15, 2019 from <http://www.cbsl.gov.lk>.
- Central Bank of Sri Lanka. (2022). National Output, Expenditure and Income. Central Bank of Sri Lanka. Retrieved on May 16, 2020 from <http://www.cbsl.gov.lk>
- Child, R. (1974). *Coconuts. Tropical Agriculture Series*, 2nd edn. Longmans Group Limited, London.
- Cochrane, W.W. (1955). Conceptualizing the supply relation in agriculture. *Journal of Farm Economics*, 37(5), 1161-1176.
- Coconut Development Authority (2021). Coconut Statistics, Coconut Development Authority, Colombo.
- Cummings, J.H. and Askari, H. (1977). Estimating Agricultural supply response with the Nerlovian model: A survey. *International Economic Review*, 18 (2), 257-292.
- Dasgupta, I. (2009). Supply Theory sans Profit Maximization. *The B.E. Journal of Theoretical Economics*, 9(1), Article 26.
- Department of Census and Statistics. (2019). *Household Income and Expenditure Survey 2016 - final report*.
- Food And Agriculture Organization of the United Nations. (2020). *FAOSTAT*, Coconuts, Production/Crops. Statistical Division. 2018. Retrieved on May 14, 2020 from <http://www.fao.org/faostat>.
- French, B. C. and Mathews, J. L., (1971). A supply response model for perennial crops, *American Journal of Agricultural Economics*, 53(3), 478-490.
- Kalaitzandonakes, N.G. and Shonkwiler, J.S. (1992). A state-space approach to perennial crop supply analysis. *American Journal of Agricultural Economics*, 74(2), 343-352.
- Karunakaran, N. and Gangadharan, K. (2014). Supply response of Coconut cultivation in Kerala. *Economic Affairs*, 59(4), 681-686.
- Mahindapala, R. and Pinto, J. L. J. G. (1991). Coconut Cultivation, Coconut Research Institute of Sri Lanka, Lunuwila.
- Leaver, R. (2004). Measuring the supply response function of Tobacco in Zimbabwe. *Agrekon*, 43(1), 113-131.
- Nerlove, M. (1958). Distributed lags and estimation of long-run supply and demand elasticities: Theoretical considerations. *Journal of Farm Economics*, 40(2), 301-311.
-

Samarajeewa, S. R., Thanaweera arachchi, P. T., Fernando, M. T. N. and Rathnasiri, N. G. S. C. (2001). Supply response analysis in coconut production. Sri Lanka Association for the advancement of science, Proceedings of the 57th Annual Session, Part 1, 265-267.

Silva, N.T. M. H. D. & Tisdell, C. A. (1981). Response of Coconuts to Fertilizer and Advice to Sri Lankan Growers: An Aggregative Approach. *Ceylon Cocon. Q.*, 32, 72- 79.

Vrankić, I. and Krpan, M. (2017). Short run profit maximization in a convex analysis framework. *Interdisciplinary Description of Complex Systems*, 15(1):1-15.

RESEARCH ARTICLE

Enhanced Coconut Somatic Embryogenesis of Unfertilized Ovary Derived Callus through Improved Media Composition

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ABSTRACT

Coconut is an economically demanding plantation crop in Sri Lanka, having an out-breeding nature with no vegetative propagation methods. Propagation by seeds generates heterogenous population and it is not sufficient to fulfill the increasing demand for coconut plantlets. Therefore, *in vitro* propagation is the ultimate approach of producing clones of coconut from the mother palms of interest. Coconut *in vitro* propagation is achieved through somatic embryogenesis and improvements of this process will easily cater the demand for plantlets. Hence, the main objective of the study was to enhance somatic embryogenic pathway of coconut using two different auxins (2,4-Dichlorophenoxyacetic acid [2,4-D] and 3,6-Dichloro-2-methoxybenzoic acid [Dicamba]) at the concentration of 300 μ M. These auxins were tested alone and in combination with calcium ionophore (A23187) and 22(S),23(S)-homobrassinolide. Translucent structures or the ear-like structures obtained from culturing of unfertilized ovaries were used for the treatments. After six weeks of the culture establishment number of embryogenic structures per one translucent structure were counted. Each treatment produced embryogenic structures but a significantly greater number of embryogenic structures were developed with 2,4-D containing media than dicamba. The results revealed the potential of using 300 μ M 2,4-D with calcium ionophore (A23187) and 22(S), 23(S)-homobrassinolide to improve the somatic embryogenic pathway of unfertilized ovary derived callus of coconut.

Keywords: Calcium ionophore, dicamba, homobrassinolide, *in vitro*, 2,4-Dichlorophenoxy acetic acid

INTRODUCTION

Coconut (*Cocos nucifera* L.) is one of the most important commercial and subsistence crop in the tropics and cultivated globally on about 12 million hectares. The demand for the coconut products around the world is growing exponentially (COGENT, 2018). Inadequate coconut production is a major challenge for the most of the coconut growing countries caused by losses due to senile plantations, pests and diseases. Hence, it is necessary to replant the plantations and increase the cultivation area to fulfill the growing demand for coconut products. Seed propagation of coconut is not sufficient for this purpose and also it produces heterogeneous populations. In order to supply good quality planting materials for coconut, *in vitro* clonal propagation is a suitable alternative (Sandoval-Cancino et al., 2016; Solís-Ramos et al., 2012).

In vitro regeneration of coconut is mainly achieved by the somatic embryogenesis. Coconut is highly recalcitrant for *in vitro* regeneration (Pérez-Núñez et al., 2009). Main difficulties encountered in coconut clonal propagation are slow response *in vitro*, high heterogeneity of the behaviour of the tissues *in vitro*, intensive browning of the tissues, high rhizogenesis ability and a low capacity for embryogenesis (Hochoy et al., 1999).

Various laboratories in the world have worked for the development of coconut micropropagation using different sources of explants and media formulations (Nguyen et al., 2015; Sáenz et al., 2016). Unfertilized ovary explant from -4 stage inflorescences is

one of the prominently used explant type for coconut micropropagation (Perera et al., 2007; 2008). Ovary explants are advantageous as they produce identical palms to the mother palm, compared to the plumule explants (Pérez-Núñez et al., 2006). They were reported to produce embryogenic callus consistently but with low frequency (Perera et al. 2007; 2009).

Commonly used auxin for coconut micropropagation is 2,4-dichlorophenoxyacetic acid (2,4-D) which is an important component in media formulations of coconut somatic embryogenesis (Sáenz et al., 1999). Dicamba (3,6-Dichloro-2-methoxybenzoic acid) is another plant growth regulator and herbicide that shares similarities in structure and activity to 2,4-D. It was used to substitute 2,4-D when inducing callus in some species such as wheat (*Triticum aestivum*) and maize (*Zea mays*), and sometimes considered as superior to 2,4-D (Carvalho et al., 1997).

Brassinosteroids are a class of growth promoting steroidal phytohormones which involved in plant growth and development (Wang et al., 2017). They promote cell division and cell differentiation and are found more active than or synergistic with, auxins such as indole acetic acid (IAA) or naphthalene acetic acid (NAA) (Brosa, 1999). Promising results were obtained by testing 22(S),23(S)-homobrassinolide on initial callus, embryogenic callus and somatic embryo formation in coconut plumule explants (Azpeitia et al., 2003).

It is assumed that calcium ionophores directly facilitate the transport of Ca^{2+} across the plasma membrane. Treatment with A23187 ionophore promoted somatic embryogenesis in wheat and coffee (*Coffea canephora*) cultures (Dedkova et al., 2000; Mahalakshmi et al., 2007; Ramakrishna et al., 2011). Treating embryogenic structure derived from coconut plumules with ionophore A23187 promoted somatic embryogenesis, and increased in the formation of embryogenic structures per callus, somatic embryos per callus and developing plantlets per callus (Rivera-Solis et al., 2018).

The present study was aimed to improve the coconut micropropagation protocol using different chemical treatments and media supplements including dicamba, calcium ionophore A23187 and 22(S),23(S)-homobrassinolide for unfertilized ovary derived calli. This was the first attempt to use above chemicals on improving somatic embryogenesis of unfertilized ovary derived calli.

2. Materials and methods

2.1. Plant material

Immature inflorescences, at the -4 stage (with 0 corresponding to the youngest opened inflorescence), of coconut cultivar Dwarf x Tall (DT), were collected to excise unfertilized ovaries. The basal part of rachilla, bearing female flowers were dissected from inflorescence and disinfected with 3% Clorox for 10 minutes and then rinsed four times with sterile distilled water. Ovaries were excised by removing the perianth parts of each female flower while submerging in a sterile

antioxidant solution (0.15 gL⁻¹ ascorbic acid, 0.1 gL⁻¹ citric acid, and 30 gL⁻¹ sucrose) under aseptic conditions. Sliced ovaries were cultured on callus induction medium.

2.2. Media preparation and culture conditions

Y₃ media formulation (Eeuwens, 1976) was used as the basal medium supplemented with 5% sucrose, 0.3% phytagel and 0.25% activated charcoal (Rivera-Solis et al., 2018). Medium pH was adjusted to 5.75 before autoclaving for 20 min at 121 °C. All the cultures were incubated in dark at 27±2°C.

Initial calli were obtained through 6–8-week incubation of cultured unfertilized ovaries (as explained previously) on callus induction medium, which is Y₃ basal medium supplemented with 600 µM 2,4-D. For the present study, about 2x2x2 mm³ translucent structures separated from the initial calli, were cultured on different media types and pre-treatments (Table1). Medium I was Y₃ supplemented with 300 µM 2,4-D while medium II was supplemented with 300 µM Dicamba. For T2, T3 and T6 translucent structures were pre cultured on medium III for three days before sub culturing on medium I or medium II. Medium III contained 1 µM 2,4-D and 1 µM 22(S),23(S)-homobrassinolide. For T1, T3, T5 and T7 cultured translucent structures were treated with 10 µL of 1µM calcium ionophore A23187 deposited with a micropipette on the explant (Rivera-Solis et al., 2018).

Table 2: Different media combinations and pre treatments

Treatment	Pre culturing medium/Pre treatment	Initiation / Sub culturing medium
Control	-	Medium I
T1	Medium I + 1 μ M Ca ionophore	-
T2	Medium III	Medium I
T3	Medium III + 1 μ M Ca ionophore	Medium I
T4	-	Medium II
T5	Medium II + 1 μ M Ca ionophore	-
T6	Medium III	Medium II
T7	Medium III + 1 μ M Ca ionophore	Medium II

2.3. Statistical analysis

The number of embryogenic structures that can be proceeded to the somatic embryogenesis per an initial translucent structure were counted from 480 replicates (60 replicates per each treatment). The data were analyzed using SAS statistical software under Poisson regression.

3. Results and Discussion

The cultured unfertilized ovaries developed initial calli (Figure 1-A) after 6 to 8 weeks. This initial calli further

developed ear-like structures (translucent structures). The treatments were started with the translucent structures (Figure 1-B) that were separated from the initial calli. The translucent structures then developed embryogenic callus (Figure 1-C) with small pearly white structures referred to as embryogenic structures (Figure 1-D). The embryogenic structures (about 3x3x3 mm³) were separated from the embryogenic callus and counted for each treatment.



Figure 1: Morphological variations during the formation of embryogenic structures from initial calli. (A) Initial calli; (B) Translucent structures separated from initial calli; (C) Embryogenic callus with embryogenic structures; (D) Embryogenic structures separated from embryogenic calli.

According to the morphological characteristics of embryogenic calli (Figure 2), visible differences were clearly observed under different treatments. With comparison to the samples cultured on media with 2,4-D (Fig. 2 C-T3), all the Dicamba incorporated media produced calli with few embryogenic structures having roots (Fig.2 T4-T7.b). Most of them generated 1-5 germinating somatic

embryos on each callus (Fig.2.T4-T7.a). Calli developed after the homobrassinolide pretreatment (medium III) produced pure white calli (productive calli) with minimal browning than the control which formed off-white calli. A different morphological characteristic was expressed by ionophore treated calli, which developed curved embryogenic calli into the main base (Fig.2-T1, T3).

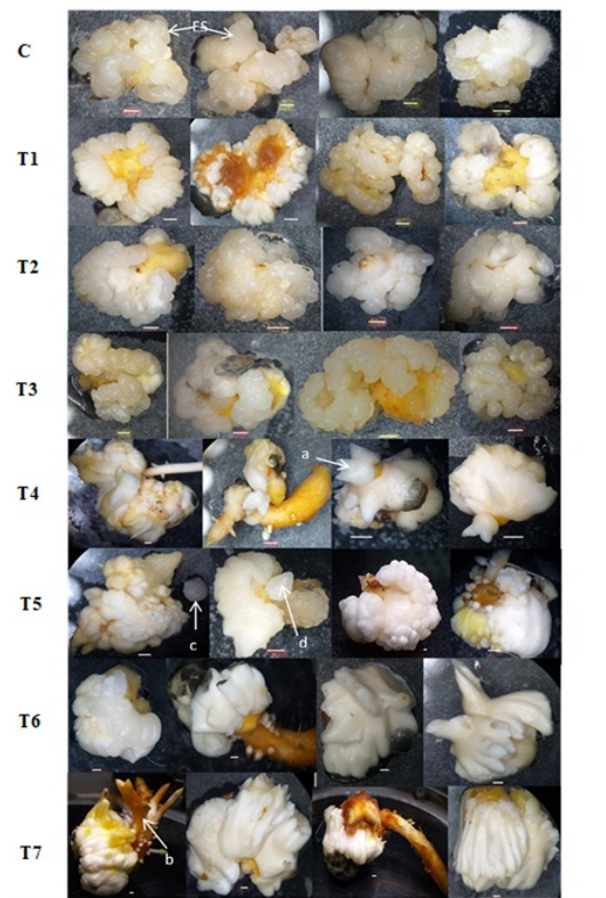


Figure 2: Morphological characteristics of calli after six weeks under different treatments (Bar = 1 mm)

Embryogenic structures (ES); germinating somatic embryos (a); roots (b); globular shaped somatic embryo (c); torpedo shaped somatic embryo (d)

The translucent structures produced varying number of embryogenic structures under each treatment (Table 2). But a significantly greater number of embryogenic structures were obtained with 2,4-D (C, T1, T2, T3) incorporated media compositions rather than dicamba (T4, T5, T6, T7).

Table 2: Mean number of embryogenic structures produced per an initial translucent structure of different treatments

Treatment	Mean number of embryogenic structures per translucent structure ($\pm$ Standard error)
C	4.33 ^a $\pm$ 0.68
T1	4.41 ^a $\pm$ 0.17
T2	4.59 ^a $\pm$ 0.28
T3	4.55 ^a $\pm$ 0.27
T4	1.70 ^b $\pm$ 0.17
T5	1.83 ^b $\pm$ 0.17
T6	1.80 ^b $\pm$ 0.17
T7	1.90 ^b $\pm$ 0.18

^{a,b} Different superscripts indicate significant difference ($P \leq 0.05$)

The media with 2,4-D and Dicamba showed significantly different results on formation of embryogenic structures. More callus formation (~4.33 fold) was observed in media with 300 μ M 2,4-D while, significantly less callus formation (~1.7-fold) was observed in media with 300 μ M Dicamba. In coconut micropropagation, 2,4-D is the prominent auxin used in many protocols (Luis et al., 2012). Dicamba was used in the presence of ascorbic acid and sorbitol during secondary somatic embryogenesis of oil palm *in vitro* propagation, which resulted in 21.55 secondary somatic embryos per one primary haustorium embryo (Te-chato & Hilae, 2007). Dicamba has given promising results in genotype independent somatic embryogenesis from immature zygotic embryos of tropical maize inbred lines (Akoyi et al., 2013). Therefore, more experiments are needed to test the effectiveness of Dicamba for somatic embryogenesis of ovary derived callus of coconut.

Calcium ionophore A23187 has been successfully utilized for somatic embryogenesis of coconut. It was reported to produce 2.8-fold embryogenic structures per callus, 1.5-fold somatic embryos per callus and 1.5-fold developing plantlets per callus. Further, it was observed that the greater expression of CnCERK (Coconut Somatic Embryogenesis Receptor Kinase) gene at initial stages of the culture period under the ionophore treatment (Rivera-Solis et al., 2018). In the present study, significant amount of embryogenic structures were obtained with calcium ionophore treatments when 2,4-D was used as the growth regulator.

The effect of the 22(S), 23(S)-homobrassinolide on initial callus, embryogenic callus and somatic embryo formation in coconut plumule explants were tested and favourable responses were obtained in previous studies. The action of brassinosteroid depends on the exposure time of the explant to brassinosteroid. Three-day

preculture period with brassinosteroid was effective than 7-day period resulting 10.8 somatic embryos per plumule explant in previous studies. The shorter exposure time was found to be more effective since the longer pre culture time might induce brassinosteroid-induced ethylene formation (Azpeitia et al., 2003). Ethylene has inhibitory effects on somatic embryogenesis using coconut zygotic embryo explants (Adkins et al., 1999). Therefore, three-day preculture period was used in the present study for homobrassinolide pretreatment for ovary derived calli but significantly different results were obtained only when they were sub cultured in 2,4-D containing media.

This was the first study of using calcium ionophore, homobrassinolide and dicamba on unfertilized ovary derived calli of coconut cultivar DT. Hence, more studies should be performed using different concentrations and pre culturing time periods to evaluate the effectiveness of these chemicals.

4. Conclusions

The results indicated morphological variations of the embryogenic calli produced by the different treatments. Pure white calli were observed with the homobrassinolide pre-culture (medium III) with minimal browning. Number of embryogenic structures were significantly greater in treatments with 2,4-D (C, T1, T2, T3) rather than with dicamba (T4, T5, T6, T7). Further studies are needed using calcium ionophore, homobrassinolide and dicamba since they were reported to produce promising results with plumule explants of coconut. The findings of the present study indicated the potential of using these treatments for somatic embryogenesis of unfertilized ovary derived callus of coconut.

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Conflicts of interest

The authors declare no conflict of interest.

REFERENCES

- Adkins, S. W., Samosir, Y. M. S., & Godwin, I. D. (1999). Control of environmental conditions and the use of polyamines can optimize the conditions for the initiation and proliferation of coconut somatic embryos. In: C. Oropeza, J. L. Verdeil G. R. Ashburner R. Cardeña & J. Santamaría (Eds.), *Current Advances in Coconut Biotechnology* (pp. 321-340). Dordrecht: Kluwer Academic Publishers.
- Akoyi, J., Mgutuu, A. J., Machuka, J., Lijsebettens, M. V., Taracha, C. & Anami, S. E. (2013). Dicamba growth regulator promotes genotype independent somatic embryogenesis from immature zygotic embryos of tropical Maize inbred lines. *Journal of Life Sciences*, 7(7), 677-689.
- Azpeitia, A., Chan J., Enz, L. & Oropeza, C. (2003). Effect of 22(S),23(S)-homobrassinolide on somatic embryogenesis in plumule explants of *Cocos nucifera* (L.) cultured *in vitro*. *Journal of Horticultural Science & Biotechnology*, 78(5), 591-596.

- Brosa, D. (1999). Biological effects of Brassinosteroids. *Critical Reviews in Biochemistry and Molecular Biology*, 34(5), 339–358.
- Carvalho, C. H. S., Bohorova, N., Bordallo, P. N., Abreu, L. L., Valicente, F. H., Bressan, W., & Paiva, E. (1997). Type II callus production and plant regeneration in tropical maize genotypes, *Plant Cell Reports*, 17(1) 73-76. <https://doi.org/10.1007/s002990050355>
- COGENT. (2018). A Global strategy for the conservation and use of coconut genetic resources, 2018-2028. (R. Bourdeix and A. Prades, compilers). Bioversity International, Montpellier, France. https://www.bioversityinternational.org/fileadmin/user_upload/Cogent_bourdeix_2017-new.pdf
- Dedkova, E. N., Sigova, A. A., & Zinchenko, V. P. (2000). Mechanism of action of calcium ionophores on intact cells: ionophore-resistant cells. *Membrane & Cell Biology*, 13, 357–368.
- Euwens, C. J. (1976). Mineral requirements for growth and callus initiation of tissue explants excised from mature coconut palms (*Cocos nucifera* L.) and cultured *in vitro*. *Physiologia Plantarum*, 36, 23-28.
- Hocher, V., Verdeil, J. L., Rival, A., & Hamon, S. (1999). Application of *in vitro* techniques to the conservation and propagation of coconut palms. In C. Oropeza, J. L. Verdeil, G. R. Ashburner, R. Cardaña, & J. M. Santamaría (Eds.), *Current Advances in Coconut Biotechnology*, *Current Plant Science and Biotechnology in Agriculture* (Vol 35.) Springer Science & Business Media. https://doi.org/10.1007/978-94-015-9283-3_19
- Luis, S., Montero, M., Teresa, P., Azpeitia, A., Andrade, T., Cordova, I., Luis, C. & Oropeza, C. (2012). Coconut (*Cocos nucifera* L.) somatic embryogenesis and related gene expression. In J. Aslam, P. S. Srivastava, & M. P. Sharma (Eds) *Somatic Embryogenesis and Gene Expression* (pp. 172-183). New Delhi: Narosa Publishing House.
- Mahalakshmi, A., Singla, B., Kurana, J. P., & Khurana, P. (2007). Role of calcium-calmodulin in auxin-induced somatic embryogenesis in leaf base cultures of wheat (*Triticum aestivum* var. HD 2329). *Plant Cell, Tissue & Organ Culture*, 88, 167–174.
- Nguyen, Q. T, Bandupriya, H. D. D., López-Villalobos, A., Sisunandar, S., Foale, M., Adkins, S. W. (2015). Tissue culture and associated biotechnological interventions for the improvement of coconut (*Cocos nucifera* L.): A review. *Planta*, 242, 1059–1076.
- Perera, P. I. P., Hocher, V., Verdeil, J. L., Doulbeau, S., Yakandawala, D. M. D. & Weerakoon, L. K. (2007). Unfertilized ovary: A novel explant for coconut (*Cocos nucifera* L.) somatic embryogenesis. *Cell Biology and Morphogenesis*, 26, 21-28. <https://doi.org/10.1007/s00299-006-0216-4>
- Perera, P. I. P., Yakandawala, D. M. D., Verdeil, J. L., Hocher, V. & Weerakoon, L. K. (2008). Somatic embryogenesis and plant regeneration from unfertilized ovary explants of coconut (*Cocos nucifera* L.). *Tropical Agricultural Research*, 20, 226-233.

- [Pérez-Núñez, M. T.](#), [Souza, R.](#), [Sáenz, L.](#), [Chan, J. L.](#), [Zúñiga-Aguilar, J. J.](#), & [Oropeza, C.](#) (2009). Detection of a *SERK*-like gene in coconut and analysis of its expression during the formation of embryogenic callus and somatic embryos. *Plant Cell Reports*, 28(1), 11-19. <https://doi.org/10.1007/s00299-008-0616-8>
- [Pérez-Núñez, M. T.](#), Chan, J. L., [Sáenz, L.](#), Gonzalez, T., Verdeil, J. L. & Oropeza, C. (2006). Improved somatic embryogenesis from *Cocos nucifera* (L.) Plumule explants, *In Vitro Cellular & Developmental Biology—Plant*, 42, 37–43. <https://doi.org/10.1079/IVP2005722>
- Rivera-Solis, G., Carbonell, L. S., Narvaez, M., Rodriguez, G. & Oropeza, C. (2018). Addition of ionophore A23187 increases the efficiency of *Cocos nucifera* somatic embryogenesis. *3 Biotech*, 8, 366. <https://doi.org/10.1007/s13205-018-1392-y>
- Sáenz, L., Chan, J. L., Souza, R., Hornung, R., Rilo, E., Verdeil, J. L., & Oropeza, C. (1999). Somatic embryogenesis and regeneration in coconut from plumular explants. In C. Oropeza, J. L., Verdeil, G. R., Ashburner, R., Cardeña, J. M. Santamaría (Eds.) *Current advances in coconut biotechnology* (pp. 309–319). Springer International Publishing: Netherlands.
- Sandoval-Cancino, G., Sáenz, L., Chan, J. L., & Oropeza, C. (2016). Improved formation of embryogenic callus from coconut immature inflorescence explants, *In Vitro Cellular & Developmental Biology—Plant*, 52, 367–378. <https://doi.org/10.1007/s11627-016-9780-7>
- Solís-Ramos, L. Y., Andrade-Torres, A., Sáenz Carbonell, L. A., Oropeza Salín, C. M., & Castaño de la Serna, E. (2012). Somatic embryogenesis in recalcitrant plants. In K. Sato (Ed.), *Embryogenesis* (pp. 597–618). Rijeka: InTech.
- Te-chato, S. & Hilae, A. (2007). High-frequency plant regeneration through secondary somatic embryogenesis in oil palm (*Elaeis guineensis* Jacq. *Var. tenera*). *Journal of Agriculture Technology*, 3(2), 345-357.
- Wang, H., Wei, Z., Li, J. & Wang, X. (2017). Brassinosteroids. In J. Li, C. Li & S.M. Smith (Eds.), *Hormone Metabolism and Signaling in Plants* (pp. 291-326). London: Academic Press.

RESEARCH ARTICLE

Financial and Economic Sustainability and Risk Resiliency of Coconut-Based Agroforestry Systems in Sri Lanka

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ABSTRACT

Coconut plantations in Sri Lanka are predominantly maintained as mono-crops and receive relatively low returns than other intensified land-use systems. However, it was evident that over 20% of coconut plantations in the country could be used more productively and profitably by incorporating Coconut Based Agroforestry Systems (CBAS). Therefore, the objective of the present study was to assess the financial and economic viability and risk resiliency of CBAS compared to coconut mono-cropping. The cost-benefit analysis was carried out to determine the financial and economic viability, and a sensitivity analysis was conducted to assess the resilience to perceived risk. Data recorded by Kurunegala Plantations Limited from 88 coconut plantations representing 15 intercropping systems from 2007 to 2019 were used for the analysis. Results of the financial analysis of selected estates revealed that almost all the intercrops except groundnut and betel were financially viable. Accordingly, the highest Net Present Value (NPV), Internal Rate of Return (IRR), and Benefit-Cost Ratio (BCR) were observed in the CBAS, including rambutan and pineapple, indicating that they are the most financially viable CBAS. According to the sensitivity analysis, almost all the perennial crop-based agroforestry systems were financially viable except for guava and pepper in the presence of the three worst scenarios tested, namely; reduction of output price by 10%, removal of fertilizer subsidy, and increase of wage rate by 15%. Notably, all the CBAS were economically viable. These results provides the information for growers and policymakers in selecting risk resilience agroforestry systems under coconut.

Keywords: Coconut based agroforestry systems, cost-benefit analysis, intercropping, perennial crops, risk resiliency

INTRODUCTION

Coconut (*Cocos nucifera* L.) is a traditional and the most widespread plantation crop in Sri Lanka, covering an area of 440,536 ha (CBSL, 2019) or nearly 20% of agricultural lands in the country (DCS, 2014). It plays a significant role in the national economy by contributing about 0.7% to the Gross Domestic Production, 2.8% to the agricultural earnings, and 5.1% to total export earnings, respectively, in 2019 (CBSL, 2019). Coconut is primarily a smallholder crop, and about 85% of holdings in the country have an average holding size of less than 8 ha. The balance is over 8 ha, categorized as estates (DCS, 2014). In Sri Lanka, most of the coconut lands are maintained as monoculture and thus, receive relatively low returns than intensified land-use systems (Peiris et al., 2003a). Therefore, to improve the productivity of the existing coconut lands, the shift from coconut monoculture to Coconut Based Agroforestry Systems (CBAS) has been recognized as one of the national priorities for developing the coconut industry since the early 1990s (Liyanage, 1994). Moreover, using criteria such as rainfall regimes and soil type, Ranatunga et al. (1988) proposed that approximately 110,000 ha of coconut lands are agronomically suitable for establishing CBAS in the country. However, despite the government efforts to provide subsidies, low-interest loans, and extension support to popularize CBAS over the past two decades, the adoption rate of CBAS in the agronomically potential area is limited to 25% (Fernando et al., 2000). Even within the large-scale growers, slow progress has been observed in adopting CBAS. Among the adopters of CBAS, most growers are limited to semi-perennial crops such as banana,

pineapple, and betel. The lifetime of these semi-perennials is approximately five years; having completed the harvest, they remove the intercrops. Hence, diversified lands again become a mono-crop of coconut. Besides, landowners are reluctant to invest in CBAS due to high labor requirements and labor costs, increasing input costs, and fluctuating output prices of products of CBAS. Fernando et al. (2000) stated that risk associated with yield and price outcomes of intercrops appears to be the crucial factors constraining the adoption of CBAS by farmers, thus emphasizing the need for exploring the influence of the above less-frequently addressed aspects on the adoption of CBAS in Sri Lanka. However, despite these limitations, it has been identified that the estate sector has more potential for the establishment of CBAS as the estate sector has organized management practices with high technical expertise. The large-scale CBAS may be described best as an entrepreneur who provides the risk capital and makes all decisions regarding the scale of operation, cropping patterns, and manages the enterprise as a commercially oriented farm unit (Ranatunga et al., 1988), and large-scale coconut growers can also diversify in many different activities of which CBAS has many advantages (Idirisinghe, 2011; Pushpakumara et al., 2011). Furthermore, it is also well accepted that crop diversification is identified as one of the most commonly used risk management strategies in agricultural systems of the world.

Therefore, this paper investigates the financial sustainability and economic viability of CBAS for short-term and long-term coconut land allocation in large-scale coconut plantations. The specific objectives of this paper are to

assess the financial and economic viability of 15 CBASs and to identify the risk resilience nature of selected CBAS in the presence of perceived risks.

2. Methodology

This study used a grower-centered financial and economic costs-benefits and risk analysis of 15 CBAS, but the Coconut Research Institute recommended 26 CBAS systems for the entire Island. However, for the analysis, we considered 15 CBAS systems suggested to the estates in the Kurunegala and Gampaha districts with suitable Agro-Ecological Zones. The activity budgeting method was employed to develop cost and benefit cash flows. Both economic and financial analysis were carried out to determine the financial and economic viability of the selected class. In addition, a sensitivity analysis was conducted to assess the risk resiliency of different systems.

In financial analysis, the analysis is done from benefits accruing to individual participants or beneficiaries of a project or firms and corporations involved that would be interested in profitability in undertaking a proposed enterprise (European Commission, 2008). The financial analysis looks at the economic returns to an individual, a private firm, or a profit-oriented government corporation. It uses market prices of goods and services. In contrast, an economic analysis that measures the benefits and costs to the country does not use market prices because they are distorted. Instead, the economic analysis uses shadow prices, which reflect the opportunity cost of goods and services or willingness to pay. In addition, the economic analysis makes adjustments to transfer payments involved in the project

accounts, such as interest payments, taxes, or subsidies. However, in financial analysis, the entities do not add to the country's existing resources and merely represent the transfer of resources. Therefore, transfer payments, such as taxes, subsidies, and interest payments, are considered (Amarasinghe, 2015).

2.1. Financial analysis

Generally, the financial analysis assesses the economic effect, overall returns, self-financing capability, investment and debt repayment capacities, overall financial projection, and financial management of agricultural projects (Gittinger, 1994). For a new project, the main objective of the economic analysis is to demonstrate that the financial cash flows expected to be generated by attractive to the prospective investors, inducing them to contribute equity funds to the particular project rather than to employ it elsewhere. For a continuing enterprise, financial success would mean the ability to generate enough revenue to meet all its financial obligations on a timely basis and command an adequate level of working capital for continued operations. It also implies the ability to generate a reasonable rate of return on capital employed. The analyses on which such judgments can be based on Internal Rate of Return (IRR), the Net Present Value (NPV), and Benefit-Cost Ratio (B/C). NPV facilitates identifying feasible projects with the scale of operation, IRR guide to ranking the projects and B/C ratio provides the idea of the resource use efficiency (Gittinger, 1994). These three indicators provide a comprehensive analysis of the coconut-based agroforestry systems (CBAS).

The NPV of a project is the sum of the discounted net flows of a project. The NPV is a very concise performance indicator of an investment project. It represents the present amount of net benefits flow generated by the investment expressed in one single value, which was calculated by using the equation (1) as follows:

$$NPV = \sum_{t=1}^n \frac{B_t - C_t}{(1+i)^t} \quad (1)$$

Bt is the benefit in the year, Ct is the cost in the year, and i is the discount rate, n is the number of years, and t = 1, 2, ..., n. Therefore, a positive NPV, (NPV > 0), means that the project generates net benefit, and it is generally desirable either in financial terms or in economic terms.

The IRR is an indicator of the relative efficiency of an investment. The formal selection criterion for the IRR is to accept all independent projects having an internal rate of return equal to or greater than the opportunity cost of capital as given in equation (2).

$$\sum_{t=1}^n \frac{B_t - C_t}{(1+i)^t} \quad (2)$$

The Benefit-Cost Ratio (BCR) is the present value of project benefits divided by the current value of project costs, as shown in equation (3).

$$BCR = \frac{\sum_{t=1}^{t=n} \frac{B_t}{(1+i)^t}}{\sum_{t=1}^{t=n} \frac{C_t}{(1+i)^t}} \quad (3)$$

2.2. Financial sustainability

One of the most critical issues to be checked is the financial sustainability of the investment, which implies that, for each year, the cumulated sum of the inflows must be higher than the outflows of that year. Therefore, the resources must be organized to satisfy this condition (European Commission, 2008). Therefore, the total Gross Margin (TGM), returns to variable cost, returns to labor, and returns to chemical fertilizer were calculated and used to assess the financial sustainability of the production systems.

Annual Total Gross Margin (TGM) estimates the sum of annual net cash flows in the CBAS (Fernando et al., 2000). The yearly gross output of each crop was multiplied by the price to define annual gross income and deducted from the variable cost to determine the annual gross margin.

2.3. Assessment of resilience to possible risk components

To assess the risk resilience, a sensitivity analysis was carried out. The three most crucial variables of coconut plantations, namely, labor wages, fertilizer subsidy, and output prices were used in the risk assessment. The current fertilizer subsidy is 48% of the cost of fertilizer which is more critical for plantation crops (Ministry of Finance, 2019). Hence, it is possible to consider the worst-case scenario of removing fertilizer subsidy. From 2007 to 2018 (The Kurunegala Plantations Limited [KPL] was taken over by the government in 2005 and started the diversification of which records are available from 2007), the average nominal wage increase of the coconut sector was approximately 15%. Hence, it is possible to

consider a worse scenario on the labor wages annual increment of 15%. According to the information during the same period, nominal output prices of the above crops varied approximately by 10%. Therefore, it is also possible to consider a worse scenario on the decrease in output prices by 10%. The Financial Discount Rate (FDR) is the opportunity cost of capital. Opportunity cost means that one will sacrifice a return for another project when using money in one project. There are several methods to calculate FDR. In this study, the actual (weighted average) cost of capital, which was 9% (CBSL, 2019), was used as the financial discount rate for the analysis. This analysis used 0.4 ha as a land unit which is equal to one acre.

2.4. Economic analysis

The economic analysis helps to design and select projects that contribute to the well-being of a country. An enterprise is interested in financial profit and the stability of that profit. At the same time, society or government is concerned with much broader objectives such as food security, rural employment, poverty alleviation, and resulting net benefits to society as a whole (Amarasinghe, 2015). In this study, a benefit-cost analysis was undertaken to measure the Economic Internal Rate of Return (EIRR), Economic Net Present Value (ENPV), and Economic Benefit-Cost ratio (EBC).

Economical prices are usually derived by adjusting financial or market prices. Thus, a sound financial analysis provides the basis for a credible economic analysis. However, several adjustments may have to be made to convert the financial costs and benefits. In this sense, data generated for financial analysis

were used as a primary source for determining economic prices. Therefore, all the costs and benefits of the research have been valued in monetary terms by converting the financial values using the Border Price Equivalent Value (BPEV) and standard conversion factors (SCF) appropriately. The detailed procedure for generating the economical prices is given in Table 2. The economical cost of the land was calculated based on the leased rental used by the company for leasing out the land; KPL has leased out the land for out-growers at LKR. 6000/- per annum with an annual growth rate of 0.2, and considered a 9% interest rate as the opportunity cost of capital. Accordingly, the economic price of the land was calculated by using equation (4) as follows:

$$V=R/(i-g) \quad (4)$$

Where;

- V = The imputed value of a parcel of land
- R = Annual rent or income from the land
- i = Interest rate or opportunity cost of capital
- g = Expected real growth rate of the rental price of land

The financial analysis assesses the financial viability of the CBAS in terms of overall financial projection and financial management of the considered agroforestry system. Financial and economic indicators are calculated for 15 CBAS consisting of annual, semi-perennial, perennial crops, and coconut mono-crop cultivation. Coconut is a perennial crop, and the CBAS observed in the field included annual, semi perennial, and perennial crops. The costs and benefits of such crops occur at different times. Therefore, NPV, IRR, and BCR were employed to compare the net worth of the monocrop and integrated system

over the entire production period. In contrast, TGM was used to measure financial sustainability.

2.5. Data

The database with unit cost tables was prepared by collecting data from 88 estates belonging to KPL. The estates are situated in all three climatic zones (the Wet, Intermediate and Dry). Since the coconut has been grown in three climatic zones in Sri Lanka, the analysis used 15 CBAS representing three climatic zones of the country. Data recorded by the KPL from 2007 to 2019 for 15 CBAS were used for the study. Details of economic lifespan and other agronomic aspects of the 15 CBAS selected for the analysis are presented in Table 1 and mono-cropped coconut plantations. The prepared unit cost tables were verified with some parallel data collected by the Coconut Research Institute, Department of Export Agriculture, and Department of Agriculture. The calculations of the annual gross margin for each crop are based on January 2019 factor and product prices. The crop budget for 15 CBAS was prepared to examine the financial viability under the possible three major risk components.

The economical prices of outputs were calculated according to border equivalent price value. The economic value of labor was obtained by converting the market value of work into the shadow wage rate. The other relevant economic prices (i.e., price of materials and machinery) were obtained by multiplying financial costs using an appropriate standard conversion factor. Finally, the monetary worth of the land was calculated based on leased rental used by the company for leasing out the land.

Inorganic fertilizer is the only imported input used in these cropping systems. Therefore, CIF fertilizer prices were obtained from fertilizer companies in dollar terms and converted into local currency using the exchange rate. Then relevant costs such as bagging and transportation were added to get the import parity price at the farm gate. A summary of the financial and economical prices used for the analysis is shown in Table 2.

Table 1 : Details on crops, their economic lifespan, and other agronomic aspects used for the analysis.

Crop	Suitable climatic Zone	Economic lifetime (years)	Unit price (LKR/kg)	Average labor requirement (md/0.4 ha /yr)	Average fertilizer requirement (kg/0.4 ha/yr)	Variety of crop	Spacing (meters)	Plants/0.4 ha as a monocrop	Plants/0.4 ha under coconut
Annual Crops									
Ginger	IZ, WZ	1	190	105	450	Local, Chinese	0.3*0.3	30,000	20,000
Ground Nut	DZ	1	125	75	340	MI-1	0.4*0.4	50,000	35,000
Semi Perennial Crops									
Banana	All Three	5	40	57	375	Local varieties	2.4*2.4	675	250
Betel	IZ	5	1.5/leaf	167	621	Local varieties	0.4*0.3	2,000	1,000
Papaw	DZ, IZ	5	30	59	294	Rather, Red Lady	3*3	450	300
Pineapple	WZ, IZ	5	115	49	739	Mauritius	0.6*0.4 Single raw	6,000	4000
Perennial Crops									
Cinnamon	WZ, IZ	20	2200	30	115	Local	0.9*1.2	3,600	1,024
Cocoa	WZ, IZ	20	400	12	150	NA 32 Millawana	2.4*3	445	200
Dragon Fruit	IZ, DZ	20	310	63	630	Red	3.6*3.6	300 Posts	150 Posts
Mango	DZ, IZ	20	200	34	65	TOMEJC	7.2*7.2 (Monocrop) 8*16 (Intercrop)	75	30
Guava	DZ, IZ	20	80	56	625	Horana	3.6*3.6 (Monocrop) 3.6* (Intercrop)	300	150
Pepper	WZ, IZ	20	650	65	338	Panyur	2.4 * 2.4	688	350
Cashew	DZ, IZ	20	300	14	56	WUSS	6*6 (Monocrop) 8*16(Intercrop)	80	30
<i>Rambutan</i> (<i>Nephelium lappaceum</i>)	WZ	20	2.5/fruit	27	79	Malwana special	10*10(Monocrop) 8*16(Monocrop)	40	30
<i>Durian</i> (<i>Durio</i>)	WZ	20	150/fruit	15	114	Golden pillow	10*10(Monocrop) 8*16(Monocrop)	40	30

Note: Wet Zone (WZ), Intermediate Zone (IZ), and Dry Zone (DZ) Fertilizer requirements and spacing of crops are based on respective institutional recommendation

Table 2 : Financial and economical prices used in the analysis.

Description (Input/ Output)	Financial Price (LKR)	Economic Price (LKR)
Output prices (LKR/Kg) FOB prices of outputs were converted into economic prices at the farm gate		
Banana	40.00	58.00
Betel	187.00	301.00
Cashew	300.00	228.00
Cinnamon	2,200.00	1468.00
Cocoa	400.00	293.00
Dragon fruit	310.00	217.00
<i>Durian</i>	150.00	153.00
Ginger	190.00	189.00
Groundnut	125.00	204.00
Guava	80.00	148.00
Papaw	30.00	40.00
Pepper	650.00	675.00
Pineapple	115.00	156.00
<i>Rambutan</i>	75.00	107.00
TOMJC mango	200.00	185.00
Price of materials (LKR/Kg) Financial price of materials (Plants, fertilizer, chemical, tools and equipment) were multiplied by 0.92 to get the economic price (Calapur irrigation project ADB, 2013)		
Banana	25,000.00	23,000.00
Betel	64,000.00	58,880.00
Cashew	6,600.00	6,072.00
Cinnamon	34,582.00	31,815.00
Cocoa	5,900.00	6,007.60
Dragon fruit	280,000.00	274,988.00
<i>Durian</i>	6,300.00	5,796.00
Ginger	99,550.00	91,586.00
Ground nut	5,760.00	5,299.00
Guava	25,125.00	28,635.00
Papaw	25,800.00	23,736.00
Pepper	19,016.00	17,495.00
Pineapple	331,500.00	327,943.20
<i>Rambutan</i>	6,400.00	5,888.00
TOMJC mango	21,000.00	19,320.00

Machinery(LKR/Tractor hour) Financial cost of machinery usage was multiplied by 0.87 to get the economic value (Jalalpur irrigation project ADB, 2013)

Banana		25,000.00	21,750.00
Betel		39,000.00	33,930.00
Dragon fruit		23,250.00	20,227.50
Ground nut		11,000.00	9,570.00
Guava		6,000.00	5,220.00
Pineapple		2,000.00	1,740.00
Labor wages (LKR/manday)	A shadow wage rate of 0.70 was used for unskilled labor	851.00	595.00
Fertilizer cost (LKR/kg)	CIF price of fertilizer is converted into local currency and multiplied by the shadow exchange rate of 1.11 and add bagging and transport charges to get import parity price at the farm gate (ADB, 2012).	30.00	70.00

(Sources: Financial prices- Sri Lanka Customs (2019), AgriStat (2019), Economical prices- Authors' calculations)

3. Results and discussion

Financial and economic indicators were calculated for 15 CBAS consisting of annual, semi-perennial, perennial crops, and coconut mono-crop cultivation. Comparisons of different agroforestry models with coconut monoculture revealed that the estimated financial and economic indicators are higher in agroforestry models, where some models are more profitable than others. Furthermore, CBASs achieve high and early economic benefits at the farm level and increase employment opportunities.

3.1. Results of the financial analysis

The financial analysis assesses the economic viability of the CBAS in terms of overall financial projection and financial management of the considered agroforestry system. Coconut is a perennial crop, and the

intercrops observed in the field included annual, semi-perennial, and perennial crops. The costs and benefits of such crops occur at different times. Therefore, NPV, IRR, and BCR were employed to compare the net worth of the monocrop and integrated system over the entire production period. In contrast, TGM returns to variable cost and returns to inputs were used to measure financial sustainability. Results of the NPV, IRR, BCR, and economic sustainability indicators are shown in Table 3.

According to the Table 3, which presents the estimated financial performance indicators for the 15 CBAS, in an annual crop-based system, only ginger can be grown profitably under the considered agronomic and climatic conditions, as it has a positive NPV with a BCR above 1 (i.e., BCR=2.1). Conversely, groundnut under coconut has a negative NPV with a BCR lower than 1, indicating that this system is financially non-viable under the

considered conditions because ginger generates more income than groundnut. Pineapple generated the highest financial returns among the four crops considered in the semi-perennial crop-based systems than banana and papaya. According to the BCR, growing pineapple was the most efficient option than other semi-perennial crop-based systems.

On the other hand, the betel under coconut was not viable in the financial analysis under the company's management because betel utilized

more labor than other semi-perennial crop-based systems. Hence, growing betel in large-scale CBAS is not financially viable mainly due to the continuous requirement of caring the crop and due to sudden price fluctuations. On the other hand, under the perennial crop-based systems, rambutan generated the highest rate of return. This is due to the high-income generation ability of rambutan from 0.4 ha compared with other perennial intercrops. According to the BCR, this is the most efficient option a grower can choose.

Table 3 : Financial indicators of annual, semi perennial and perennial CBAS.

Crop	Achieving financial sustainability (years)	IRR	NPV (LKR)	BCR
Annual Crops				
Ginger	1		908,603	2.10
Groundnut	-		-52,286	0.84
Semi-Perennial Crops				
Betel	-	-26%	-76961	0.95
Papaw	2	37%	35,636	1.14
Banana	2	33%	41,911	1.13
Pineapple	2	58%	494,377	1.76
Perennial Crops				
Guava	5	15%	143,584	1.19
Cashew	4	24%	84,460	1.62
<i>Durian</i>	8	31%	237,293	2.82
Pepper	7	18%	203,242	1.36
Cocoa	5	35%	253,803	2.45
Cinnamon	6	19%	214,516	1.76
Mango	4	39%	373,858	2.29
Dragon Fruit	3	25%	787,211	1.84
<i>Rambutan</i>	6	44%	526,000	3.04

Recently introduced dragon fruit-based systems generated satisfactory financial returns. Even though mango generated a higher rate of return than cinnamon, BCR indicated that growing mango is more resource-efficient than growing cinnamon. Other perennial crops such as cocoa, pepper, durian, cashew, and guava also generated favorable financial returns.

Different CBAS also reached financial sustainability at other times. This can be attributed to the substantial reduction of the payback period in CBAS due to the generation of multiple income sources even when coconut produces zero income until its bearing stage (Peiris et al., 2003a; 2003b). Moreover, agroforestry practices ensure a proper cash flow during the pre-bearing period (Liyanage et al., 1986). For example, annual crop-based systems reached a positive gross margin within the same year of establishment, whereas semi-perennial crop-based plans got financial sustainability within two years. On the other hand, perennial crop-based systems take 4 to 8 years to reach financial sustainability. Dragon fruit-based cropping systems have the lowest TGM, thus attaining financial sustainability within the shortest period (i.e., four years). In contrast, Durian-based systems take eight years to reach financial sustainability, which is the highest among the perennial crop systems.

3.2. Total gross margin

Annual input and output data (cost of production and net sale average) of coconut monoculture and intercropping were used to calculate annual TGM. The length of the project period used for calculating TGM for annual and semi-perennial intercrops was one year and five years, respectively. Although we

considered the lengths of the coconut-based perennial agroforestry systems for the overall analysis as 20 years, it generates positive constant returns after 5 to 9 years. Therefore, we have included TGM only for nine years instead of 20 years to avoid repeating the exact figures up to 20 years.

Table 4 shows the annual TGM of annual and semi-perennial intercropping systems in comparison with mono-cropping coconut. Tables 5 and 6 show the yearly gross margin of perennial intercropping systems in contrast with mono-cropping coconut. In general, coconut monoculture generates negative TGM during the initial years after planting till the yield is established due to its high initial investment. However, in the analysis, an already existing mature coconut monoculture system was considered; thus, the long-term's establishment costs for mature coconut plantations can be regarded as historical costs. Besides, coconut monoculture stands considered in the analysis comprised of Sri Lanka Tall (SLT) palms. The nut yield of SLT increases progressively every year after initial bearing until a maximum is attained at about 16-18 years and is maintained after that, depending on the environmental conditions (Liyanage and Gunathilake et al., 1998). Since the average commercial age of the representative monocrop used in the analysis is 50 years, it has been assumed throughout the study that the survey sample has already achieved the maximum yield. Further, the present average production of 50 nuts/tree would be expected to continue during each year of the entire five years considered in the analysis.

According to the Table 4, all the annual and semi-perennial intercrops, except groundnut, generated higher positive TGM than coconut

monocrop under the estate management. Among the two annual intercrops considered, ginger gives higher TGM than that of coconut monocrop. Pineapple's cropping systems have negative gross margins in the first year in the semi-perennial intercrop category. Pineapple does not generate returns in the first year but incurs high establishment costs. However,

from the second year onwards, it commences producing much higher gross margins than the monocrop system. CBAS consisting of banana, papaw, and betel generate higher positive gross margins than monocrop system in the second year ahead, more elevated than coconut monoculture.

Table 4 : Gross margins of annual and semi perennial intercrops.

Gross Margin (LKR./0.4 ha)					
System	1styear	2ndyear	3rdyear	4thyear	5thyear
Coconut Monocrop	23,137	23,137	23,137	23,137	23,137
Annual Crops					
Coconut +Groundnut	1,491	12,227	12,227	12,227	12,227
Coconut + Ginger	256,732	256,732	256,732	256,732	256,732
Semi Perennial Crops					
Coconut + Pineapple	-442,276	359,074	368,503	337,073	253,198
Coconut + Banana	-79,812	73,135	114,370	42,370	38,837
Coconut +Papaw	-42,699	51,041	83,914	41,265	42,796
Coconut + Betel	-111,745	41,440	39,542	35,283	38,687

According to Table 5 and Table 6, perennial intercrops with coconut take 3 to 5 years to produce a positive gross margin and achieve an annual TGM higher than coconut monocropping in varying years. However, among

the perennials, cropping systems which include dragon fruit, generate positive TGM within three years and continue to achieve the highest TGM among the perennials after five years.

Table 5 : Gross margin of perennial intercrops (LKR/0.4 ha).

Year	Coconut monocrop	Coconut+ Dragon fruit	Coconut+ Guava	Coconut+ Pepper	Coconut+ Cinnamon
1	23,137	-354,328	-92,636	-61,624	-64,854
2	23,137	-27,091	-50,714	-31,054	1,072
3	23,137	22,603	-67,203	-32,340	-1,015
4	23,137	67,489	2,883	8,127	-1,266
5	23,137	218,571	47,692	39,693	-133
6	23,137	217,869	95,692	48,686	72
7	23,137	216,869	95,692	76,033	106,489
8	23,137	217,869	95,692	102,529	105,638
9	23,137	216,869	95,692	102,529	105,638

Table 6 : Gross margin of perennial intercrops (LKR/0.4 ha).

Year	Coconut monocrop	Coconut + Rambutan	Coconut+ Durian	Coconut+ Cashew	Coconut+ Cocoa	Coconut+ Mango
1	23,137	3,895	7,150	-671	-8,072	-27,184
2	23,137	14,069	18,472	17,217	11,909	10,495
3	23,137	12,738	17,560	16,752	10,578	9,730
4	23,137	15,157	16,247	21,118	17,378	37,989
5	23,137	11,821	13,555	27,095	36,425	56,847
6	23,137	35,935	11,553	32,261	53,625	79,013
7	23,137	60,829	18,041	34,410	74,025	104,758
8	23,137	96,627	28,563	42,011	84,425	104,758
9	23,137	131,574	39,486	42,011	84,425	104,758

In summary, the annual gross margin analysis suggests that all the intercropping systems considered are superior to monocrop systems in terms of margins /unit of land. Even though net income obtained from a unit holding under coconut is the lowest among plantation crops grown in Sri Lanka (Liyanage, 1994), CBAS can provide a high and consistent income than monocropping (Liyanage et al., 1986; Ranathunge et al., 1988; Gunathilake and Liyanage, 1996;). Reducing fertilizer, weeding, and inter-cultivation costs for coconut has been identified as other secondary benefits. Furthermore, Abeygunawardena and Fernando (1992) have also shown that agroforestry can substantially raise the monetary value of the coconut lands, where mean land value has been increased (8-115%) as a result of coconut intercropping compared to coconut monocropping. Among the annual, semi-perennial, and perennial systems, ginger, pineapple, and dragon fruit are more attractive to provide positive annual gross returns during the entire planting period. Ginger generates comparatively high TGM. Thus, such a crop is much more suitable for poorly endowed farmers and smallholders concerned with sustaining a positive annual cash flow, no matter how low, rather than waiting longer to obtain higher cash flows. In pineapple, the initial investment is much higher than that of other crops in the category. The crop generates its highest TGM five years after the field establishment in dragon fruit. Thus, these crops may be suitable for better-endowed farmers as they can bear the high initial cost and be able to wait for higher returns occurring at later stages (Fernando et al., 2000).

3.3. Assessment of resilience to possible risk components-sensitivity analysis

Results of the sensitivity analysis against the baseline are given in Table 7 and Table 8. The sensitivity analysis for the CBAS was carried out considering seven different scenarios as follows;

- i. Scenario 1: Output price decreased by 10%
- ii. Scenario 2: Wage rate increased by 15% (LKR. 979.00 labor/day)
- iii. Scenario 3: Removal of fertilizer subsidy (LKR. 70.00/kg)
- iv. Scenario 4: Output price decreased by 10%, with the wage rate increased by 15% (LKR. 979.00)
- v. Scenario 5: Wage rate increased by 15% (LKR. 979.00) with the removal of fertilizer subsidy (LKR 70.00)
- vi. Scenario 6: Output price decreased by 10% with the removal of fertilizer subsidy. (LKR. 70.00)
- vii. Scenario 7: Output price decreased by 10%, the wage rate increased by 15% (LKR. 979.00), removal of fertilizer subsidy (LKR. 70.00)

Results of the sensitivity analysis showed that financial viability is different in different agroforestry systems based on various risk components. According to Table 7, in ginger crops included agroforestry systems, the three indicators considered in the sensitivity analysis IRR, NPV, and BCR exhibit a marginal decrease in all seven scenarios compared to the baseline scenario. It is important to note that the NPV reduced from LKR 908,603.00 to 838,589.00, recording a marginal decline when the fertilizer subsidy

was removed (i.e., scenario 3). Moreover, the system remains financially viable even when these risk scenarios were imposed. On the other hand, the cropping system, which includes groundnut, was not economically feasible in the baseline scenario. Thus, the situation has also worsened in the seven considered methods.

In the semi-perennial intercrop systems, papaya and banana showed a significant decrease in all seven scenarios when considering the IRR. The NPV in the sensitivity analysis of papaya varied from LKR 35,636.00 to LKR -67,961.00, recordings a drastic change in the system's financial viability when all three restrictions were imposed. However, according to BCR, the economic viability of the system declines marginally. Similar movement was observed in the banana-based agroforestry system's

IRR, NPV, and BCR. In contrast, NPV is reduced slightly and still financially viable in the pineapple intercropping system, even if all three risk components are considered together. However, the BCR remains more than one in all scenarios indicating that the pineapple-based systems are financially viable within the risk components.

According to Table 8, all the perennial intercrop systems except guava and pepper, and all the other models are financially viable under the three scenarios. This indicates that the bearing of risk is higher with coconut-based perennial agroforestry systems. These systems require less labor and relatively less chemical fertilizer for a unit of land area. Due to the low labor and the less amount of fertilizer requirement, high financial returns from a unit area of land.

Table 7 : Results of the sensitivity analysis against the baseline of annual and semi perennial CBAS.

Crop	Performance indicator	Baseline	OP						
			Output price (OP) decreased by 10%	The wage rate (WR) increased by 15%	Fertilizer Subsidy (FS)=0	decreased by 10% and WR increased by 15%	WR increased by 15% and FS=0	OP decreased by 10%	OP decreased by 10%, WR increased by 15% and FS=0
Annual crop-based agroforestry systems									
Ginger	IRR								
	NPV	908,603	734,930	856,326	838,589	682,653	786,312	664,916	612,640
	BCR	2.10	1.89	1.97	1.93	1.78	1.83	1.74	1.64
Ground nut	IRR								
	NPV	-52,286	-79,076	-89,590	-105,185	-116,380	-142,489	-131,975	-169,279
	BCR	0.84	0.75	0.75	0.72	0.67	0.65	0.65	0.59
Semi perennial crops based agroforestry systems									
Papaya	IRR	37%	14%	14%	1%	-12%	-27%	-31%	
	NPV	35,636	6,386	6,864	-9,939	-22,386	-38,711	-39,189	-67,961
	BCR	1.14	1.02	1.02	0.97	0.92	0.88	0.87	0.79
Banana	IRR	33%	15%	17%	-1%	-3%	-23%	-31%	
	NPV	41,911	9,257	13,698	-16,433	-18,957	-44,647	-49,088	-77,302
	BCR	1.13	1.03	1.04	0.96	0.94	0.89	0.87	0.81
Pineapple	IRR	58%	48%	55%	45%	45%	42%	35%	32%
	NPV	494,377	383,555	469,218	377,953	358,396	352,794	267,131	241,972
	BCR	1.76	1.59	1.70	1.49	1.53	1.45	1.35	1.31
Betel	IRR	-26%							
	NPV	-76961	-214043	-238155	-170179	-375237	-331373	-307261	-468455
	BCR	0.95	0.85	0.85	0.89	0.77	0.81	0.80	0.72

Table 8 : Results of the sensitivity analysis against the baseline of perennial CBAS.

Crop	Performance indicator	Baseline	Output price (UP) decreased by 10%	The wage rate (WR) increased by 15%	Fertilizer Subsidy (FS)=0	OP		OP	
						decreased by 10% and WR increased by 15%	WR increased by 15% and FS=0	decreased by 10% and FS=0	decreased by 10%, WR increased by 15% and FS=0
Guava	IRR	15%	11%	12%	6%	8%	3%	1%	-2%
	NPV	143,584	42,671	79,285	-76,382	-21,628	-140,681	-177,295	-241,594
	BCR	1.19	1.07	1.10	0.94	0.99	0.89	0.85	0.80
Cashew	IRR	24%	21%	21%	20%	18%	17%	17%	14%
	NPV	84,460	61,693	69,108	64,993	46,341	49,641	42,226	26,874
	BCR	1.62	1.46	1.47	1.43	1.32	1.31	1.29	1.18
Durian	IRR	31%	29%	29%	26%	27%	25%	24%	23%
	NPV	237,293	199,715	223,206	201,190	185,628	187,104	163,612	149,525
	BCR	2.82	2.54	2.56	2.24	2.30	2.07	2.01	1.86
Pepper	IRR	18%	15%	15%	13%	11%	10%	9%	6%
	NPV	203,242	122,061	129,693	86,744	48,512	13,195	5,564	-67,986
	BCR	1.36	1.22	1.21	1.14	1.09	1.04	1.03	0.93
Cocoa	IRR	35%	31%	32%	28%	29%	26%	25%	23%
	NPV	253,803	210,271	238,785	200,059	195,253	185,041	156,527	141,508
	BCR	2.45	2.21	2.26	1.89	2.04	1.78	1.70	1.60
Cinnamon	IRR	19%	18%	17%	17%	15%	15%	15%	13%
	NPV	214,516	164,836	178,923	174,518	129,243	138,924	124,838	89,245
	BCR	1.76	1.59	1.58	1.55	1.42	1.41	1.41	1.27
Mango : Tom EJC)	IRR	39%	35%	35%	36%	32%	33%	33%	29%
	NPV	373,858	306,755	338,681.17	348,008	271,578	312,831	280,905	245,728
	BCR	2.29	2.06	2.05	2.10	1.84	1.90	1.89	1.71
Dragon Fruit	IRR	25%	22%	23%	20%	20%	18%	17%	15%
	NPV	787,211	610,698	717,759	557,172	541,246	487,719	380,659	311,206
	BCR	1.84	1.66	1.72	1.49	1.55	1.41	1.34	1.27
Rambutan	IRR	44%	41%	41%	41%	38%	39%	38%	35%
	NPV	526,000	446,672	499,855	500,540	420,527	474,395	421,212	395,066
	BCR	3.04	2.73	2.77	2.77	2.49	2.55	2.50	2.29

3.4. Results of the economic analysis

An economic benefit-cost analysis was carried out to estimate the EIRR, ENPV, and EBC to measure the economic viability of the CBAS. The estimated performance indicators are presented in Table 9 along with financial performance indicators. Notably, according to the estimated economic indicators, all the CBAS are economically viable. The highest ENPV and EBCR values are given by ginger, pineapple, and rambutan crops in the annual, semi perennial, and perennial cropping

systems. Remarkably, groundnut and betel crops that exhibit negative NPV in the financial analysis have positive ENPV. On the one hand, on a small scale, betel and groundnut crops with coconut may utilize more family labor and generate additional income. However, on the other hand, the existing betel market for export in the country is monopolistic. Thus financial prices are more distorted. Therefore, the economic indicators reveal the actual value of these systems, which hold for the society.

Table 9 : Economic and financial indicators of annual, semi perennial and perennial CBAS.

Crop	Achievement of						
	Financial Sustainability (years)	IRR	EIRR	NPV	ENPV	BCR	EBCR
Annual Crops							
Ginger	1			908,603	943,934	2.10	2.20
Groundnut	-			-52,286	134,719	0.84	1.44
Semi-Perennial Crops							
Betel	-	-26%		-76961	966,618	0.95	1.78
Papaw	2	37%	86%	35,636	121,559	1.14	1.44
Banana	2	33%	101%	41,911	194,264	1.13	1.62
Pineapple	2	58%	111%	494,377	976,035	1.76	2.72
Perennial Crops							
Guava	5	15%	36%	143,584	954,580	1.19	2.00
Cashew	4	24%	15%	84,460	26,945	1.62	1.21
<i>Durian</i>	8	31%	30%	237,293	237,852	2.82	2.73
Pepper	7	18%	22%	148,289	274,171	1.27	1.51
Cocoa	5	35%	31%	220,066	168,842	2.26	2.16
Cinnamon	6	19%	14%	454,560	75,335	2.57	1.31
Mango	4	39%	37%	457,737	326,519	2.57	2.13
Dragon Fruit	3	25%	11%	425,101	83,063	1.34	1.09
<i>Rambutan</i>	6	44%	58%	526,000	892,321	3.04	4.81

The study revealed that the estimated financial and economic indicators are higher in agroforestry models, and some models are more profitable than others. Furthermore, CBAS achieves high and early economic benefits at the farm level and increases employment opportunities.

4. CONCLUSIONS AND POLICY IMPLICATIONS

4.1. Conclusion

This paper assesses the adaptability of 15 CBAS by using financial and economic performance indicators. Also, the risk resilience of CBAS was evaluated in the presence of three major risk components using a sensitivity analysis. Annual time series data expanded over 13 years of 88 commercially managed estates were used for the analysis.

The results obtained for financial and economic indicators concerning coconut monocropping and CBAS provide strong evidence to prove that CBAS are financially and economically advantageous vis-à-vis coconut monocropping. Among the considered 15 CBAS, except groundnut and betel, all the other CBAS are financially profitable at the current average price levels and agronomic and management practices. According to the economic analysis, all the CBAS generate economic benefits over the investment incurred.

The sensitivity analysis results to assess the risk resiliency of 15 CBAS revealed that these cropping systems respond to the risks enacted in varying degrees. Among annual crop-based systems, only ginger shows a marginal decrease in NPV and BCR in all seven scenarios compared to the baseline, thus remaining financially viable even though these risk scenarios were imposed. Pineapple is most risk resilient in the semi-perennial crop-based systems while remaining financially profitable. In the perennial crop-based systems, almost all the perennial crops were financially viable except guava and pepper under the three worst scenarios tested, namely; reduction of output price by 10%, removal of fertilizer subsidy, and increase of wage rate by 15%. Rambutan is the most risk resilience crop. This indicates that the bearing of risk is higher with perennial crop-based agroforestry systems as these systems require less labor and a relatively less amount of chemical fertilizer for a unit area of land. Therefore, these systems are suitable for medium to large-scale coconut plantations. This allows highlighting some important aspects of the domestic policy framework to encourage growing more perennial intercrops in coconut lands and thereby promoting agroforestry systems which in return brings more financial and economic benefits to the growers and environmental benefits to the system and the country. This paper contributes to the existing literature in three ways. First, this is the pioneering attempt at applying a detailed complete economic analysis for the CBAS in the plantation sector in Sri Lanka. Second, this study compares the financial analysis's estimated results, incorporating the possible risk components and underlining the implications. Third; it compared the revenue of CBAS with coconut monoculture systems of annual, semi-perennial, and perennial CBAS.

4.2. Policy Implications

The findings of this study offer substantial insight into the empirical assessment of financial and economic viability and resilience to the possible risk at the estate level, which is a topic full of practical consequences for policymaking. The profitability of agricultural innovation is one of the key considerations for its adaption although it is not the sole criterion. This study has concluded that the CBAS generates higher financial and economic returns than that of coconut mono-crop. Thus, the low adaptability of CBAS may lie with other factors rather than the conventionally held view of the low profitability of CBAS. Therefore, implementing appropriate strategies to realize other important factors such as supplementing the demand for the management of crops including procuring of inputs (i.e. disease-free planting materials, flowering hormones, mulching materials, etc.) and providing skilled knowledge including different agronomic practices (i.e. selection of quality planting materials, use of recommended spacing and pruning techniques, timely application of fertilizer, hormones, and irrigation, timely harvesting, pest and disease management, etc.). Moreover, this study addresses the other crucial factor which seems to minimize the adaption of CBAS, which is the perceived risk associated with output and input prices. The results of the sensitivity analysis revealed that even with the perceived risk of reducing the output price by 10 %, increasing of wage rate by 15 %, and more importantly removal of fertilizer subsidy, most of the CBAS remain financially profitable. Among the perennials, risk resiliency is higher than that of the other systems. Therefore, policy measures can be implemented to derive crop-specific incentive schemes such as the provision of high-quality planting materials and providing technical

assistance. Moreover, it is crucial to revisit the effectiveness of the existing fertilizer subsidy program which constitutes a considerable portion of the state's expenditure and to take measures to either reduce or remove the existing fertilizer subsidy scheme, as the results of this study illustrated and proved the profitability of the CBAS can be maintained even with the removal of the fertilizer subsidy. Our article, therefore, shows that identifying risk resilience intercropping systems under coconut constitutes a first step in the design of policy measures aimed at boosting sustainability at the estate level through coconut-based agroforestry systems.

Declaration of Conflict of Interests

The authors declared no potential conflicts of interest concerning the research, authorship, and publication of this article

REFERENCES

- Abeygunawardena.P & Fernando, M.T.N (1992). Sustainability and non-market components in coconut-based farming systems in Sri Lanka. An economic Evaluation Paper was Presented at the 2nd Asian Farming System Symposium held Colombo, Sri Lanka, 2-5 November..
- ADB (2013). Jalalpur Irrigation Project Asian Development Bank, Manila, Philippines
- ADB (2012). Additional Financing: Dry Zone Urban Water and Sanitation Project in the Democratic Socialist Republic of Sri Lanka, Manila.
- Agstat (2019). Agricultural Statistics. Volume: XVI. Socio-Economics and Planning Centre, Department of Agriculture, Peradeniya, Sri Lanka.
- Amarasinghe, N. (2015). Design, appraisal, and sustainable development projects. Books on Demand Philippines, Inc, Manila, Philippines
- CBSL (2019). Monetary Policy Review: No.05-2019. Central Bank of Sri Lanka, Colombo, Sri Lanka. 6
- DCS (2014). Sri Lanka Agricultural Census. Department of Census and Statistics, Colombo, Sri Lanka.
- European Commission, (2008). Directorate General Regional Policy Guide to Cost Benefit Analysis of Investment Projects. European Commission. Regional and Urban Policy Communication Unit, B-1049, Brussels.
- Fernando, M.T.N, Daw ME, & Edward IE (2000). Adaption of coconut-based intercropping systems in Sri Lanka: the fallacy of conventional wisdom on economic profitability. CORD 16: 1-36.
- Gittinger, J.P (1994). Economic analysis of the agricultural project. 2nd ed. Johns Hopkins University Press, Baltimore.
- Gunathilake, H.A.J & Liyanage, M.de S (1996). Terminal Report on coconut intercropping. Council for Agriculture Research Policy, Colombo, Sri Lanka.
- Idirisinghe, I.M.S.K (2011). Socio-Economics Aspects of Coconut-Based agroforestry Systems in Sri Lanka. In: Pushpakumara DKN, Gunasena HPM, Gunathilake HAJ

- & Singh VP (eds). Increasing Coconut Land Productivity through Agroforestry Interventions. Proceedings of a Symposium Held at the Coconut Research Institute, Lunuwila, Sri Lanka on 15th March 2011. Coconut Research Institute, Lunuwila, Sri Lanka, World Agroforestry Centre, South Asia, New Delhi, India. pp. 77-86.
- Liyanage, M de S, Tejwani, K.G & Nair, P.K.R (1986). Intercropping under coconut in Sri Lanka. *COCOS* 4: 23-34.
- Liyanage, M. de S (1994). Coconut-based agroforestry in Sri Lanka. Peekay Tree Crop Development Foundation, Cochin 682020. Pp. 105-122.
- Liyanage, M. de S & Gunathilake, H.A.J (1998). Agronomy and farming systems research in coconut: Sri Lanka experiences. Proceedings of the International Cashew and Coconut Conference held at Dar es Salaam: 411-417.
- Peiris, W.B.K, Fernando, M.T.N, Hitinayake, H.M.G.S.B, Dissanayake, K.B, Gunathilake, H.A.J and Subasinghe, S.D.J.N (2003a) Economic feasibility and biological productivity of coconut-based agroforestry models in Sri Lanka. *COCOS* 15: 38-52.
- Peiris, W.B.K, Fernando, M.T.N, Hitinayake, H.M.G.S.B, Dissanayake, K.B, Gunathilake, H.A.J, and Subasinghe, S.D.J.N (2003b). Social acceptability of coconut-based agroforestry models developed for smallholders in Sri Lanka. *COCOS* 15: 60-81.
- Pushpakumara, D.K.N.G, Gunasena, H.P.M and Gunathilake, H.A.J (2011). Review of coconut-based agroforestry systems in Sri Lanka. In: Pushpakumara D.K.N.G, Gunasena H.P.M, Gunathilake H.A.J & Singh VP (eds). Increasing Coconut Land Productivity through Agroforestry Interventions. Proceedings of a Symposium Held at the Coconut Research Institute, Lunuwila, Sri Lanka on 15th March 2011. Coconut Research Institute, Lunuwila, Sri Lanka, World Agroforestry Centre, South Asia, New Delhi, India. pp. 1-28.
- Ranatunga, A.S, Liyanage, L.V.K, and Perera, R.A.J.R (1988). Coconut-based intercropping systems in the wet and wet intermediate zones. Coconut Research Institute, Sri Lanka.
- SLC (2019). Sri Lanka Customs Statistics- 2019, Sri Lanka Customs, Colombo, Sri Lanka.
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RESEARCH ARTICLE

Nutritional Properties of De-fatted Coconut (*Cocos nucifera* L.) Testa Flour Incorporated Sri Lankan Traditional Food: Roti

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ABSTRACT

Coconut testa (brown outer skin of coconut kernel) is a by-product of coconut industry and it does not have proper usage at present. The aim of this research was to evaluate the potential of coconut testa flour (TF) to substitute for wheat flour in preparation of unleavened bread or roti. TF is the ground defatted dehydrated coconut parings obtained from paring oil extraction. The dehydrated brown testa was subjected to oil extraction using cold press expeller followed by grinding into a fine flour. The quality of raw material (testa and TF) was evaluated. The samples of roti were prepared by incorporating 0%, 10%, 20% and 30% of TF instead of wheat flour and the sensory attributes of the roti were evaluated. Functional and proximate composition of flour blends and selected roti samples were compared with a control (100% wheat flour). Fat and moisture contents of dehydrated parings were 58.06% and 4.04% respectively, and the concentrations were reduced to 16.97% and 2.56% respectively while producing TF. All sensory attributes (taste, texture, appearance, smell, overall acceptability) of the TF incorporated roti were significantly different ($P < 0.05$) compared to the roti made with 100% wheat flour (WF). According to the results, TF can be successfully substituted for wheat flour up to a level of 20% to enhance nutritional qualities of roti. No significant difference of swelling capacity, bulk density, tapped density and water absorption capacity ($P > 0.05$) between the selected flour blend sample (20% TF: 80% WF) and 100% WF sample was observed. However, crude fiber and protein contents of the 20% TF and 80% WF blend were significantly ($P < 0.05$) higher than the WF only samples. The TF incorporation (20%) contributed to increase the crude fiber content of roti from 1.81% to 7.04% and protein content from 12.43% to 16.06%.

Keywords: coconut testa flour, wheat flour, unleavened bread, sensory characteristics, functional properties

INTRODUCTION

Sri Lankans consume rice and wheat flour-based food as staple food and higher percentage of per capita consumption of wheat flour is recorded among the urban and estate communities compared to the other communities (Jayatissa et al., 2014). The roti (unleavened bread made with wheat flour and shredded coconut) is a traditional food item consumed for breakfast or dinner. However, excessive consumption of wheat flour containing food causes many health hazards such as obesity, heart disease, digestive problems and hyper-insulin response due to the high gluten and starch contents (Ranasinghe et al., 2015). Moreover, there is a direct correlation of refined grains and fast-food patterns with type 2 diabetic among developing countries (Sami et al., 2017). Therefore, diabetic patients are forced to follow a strict gluten-free or low gluten and low starchy diet. Moreover, gluten-sensitive consumers demand for nutritious, low gluten food products (Oniszcuk et al., 2019).

Studies have been conducted to identify potential functional ingredients with low Glycemic Index (GI) values to replace wheat flour. Researchers reported that rice flour, soy flour and kurakkan (*Eleusine coracana*) flour can be substituted in wheat flour based - traditional foods. (Perera et al., 2015, Thathvasuthan et al., 2007). However, there are no evidences of using de-fatted coconut testa flour (TF) as a partial substitute for wheat flour-based food.

Coconut (*Cocos nucifera* L.) is a vital food crop which is a major component of the average Sri Lankan diets and an important crop in the economy of Sri Lanka. It is used for a variety of products which is exported to various countries. Desiccated coconut (DC),

Virgin coconut oil (VCO) and Coconut milk are the major processing industries which has a good export potential. Common practice of producing these products is the removal of brown skin or testa of coconut kernel. The brown skin is rich in oil with high oleic acid content compared to the oil of the white kernel.

The testa is used to extract coconut testa oil and the by-product (press cake) is rich in fiber and other nutritional ingredients and suitable for human consumption if processed with hygienic conditions. Appaiah et al., (2014) revealed that whole coconut kernel and coconut testa contain similar proximate composition with slight deviation. The TF has protein 23.82%, fat 10.12 % and carbohydrates 59.21% (Marasinghe et al, 2019). Functional and chemical composition suggest TF can be partially incorporated in place of wheat or rice flour in food preparation.

Coconut TF was produced from testa available in processing industries by sequential removal of water and fat followed by grinding. The bioactive compounds were sequentially extracted using hexane, ethyl acetate and methanol. Methanol extract of all cultivar displayed presence of phenolic compounds and flavonoid compounds such as caffeic acid, chlorogenic acid, gallic acid, ellagic acid, ferulic acid, p-coumaric acid, sinapic acid, vanillic acid, epigallocatechin, gallates, quercetin and rutin. In addition to the presence of above bioactive compounds, it also showed antioxidant and alfa- amylase inhibitory activity (Gunarathne et al, 2022). Extruding of coconut paring flour and wheat flour is possible to make string hopper which is a wheat flour or rice flour based-based extruded product (Rushdha et al, 2022). These reports indicate the suitability of coconut TF for partial substitution of coconut TF for wheat flour.

Therefore, the focus of this research is to find the best level of incorporation of TF to wheat flour for popular unleavened food “roti” to increase the health status of the diet than the roti prepared with wheat flour.

2. Methodology

2.1. Location and sample collection

The research was conducted at the Coconut Processing Research Division (CPRD) of Coconut Research Institute, Lunuwila. Fresh coconut testa was obtained from Desiccated Coconut (DC) mill at Katana, Sri Lanka. All-purpose wheat flour was purchased from a local market.

2.2. Location and sample collection

The fresh coconut testa was washed with clean water and drained. Then, it was dried at 70 °C using a cabinet-type dehydrator (Unitex Engineers, Sri Lanka) until moisture content of the testa reached to 5%. Then, oil was extracted using screw press baby oil expeller (Ashoka industries, Sri Lanka) and defatted brown color press cake was collected and ground into a fine powder using a pulverizing machine (FRITSCHPUL-14 Sri Lanka) and the flour was called defatted TF.

2.3. Identification of the best level of defatted TF for *roti* preparation

A portion of wheat flour was replaced by TF to formulate TF incorporated roti.

Wheat flour, coconut meat and table salt were mixed and water was added to make consistent dough which is kneadable and non-sticky. The dough was flattened to a uniform thickness on a banana leaf and cut into round 15 cm diameter and 5 mm thickness and baked on a flat clay pan using a gas cooker. Once a side of the roti was baked for about 3 minutes, it was turned over to bake for another 2 minutes.

2.4. Data collection

2.4.1. Quality of raw materials

The initial quality of the raw material (testa) was analyzed to determine the free fatty acid content (FFA) and peroxide value (SLSI 2002). The proximate composition of the dehydrated testa and the TF were analyzed for moisture content (AOAC method 990.20: 1999), fat content (Randle method: 945.16), protein content (Kjeldahl method 955.04), and ash (dry ash: 900.02A) content

2.4.2. Evaluation of sensory properties of TF incorporated *roti*

Sensory evaluation was conducted to determine the appearance, odour, taste, texture and overall acceptability of *roti* samples using 25 numbers of semi trained panelists. The panel consisted of females and males of 20 - 50 years of age. The panelists were given written instructions and were explained the purpose of the

Table 1 : Recipes for TF based roti formulation

Treatment	Wheat flour (g)	Testa flour (g)	Coconut meat (g)	Water (ml)	Salt (g)
Control	100	-	60	30	3.5
R ₁	90	10	60	30	3.5
R ₂	80	20	60	30	3.5
R ₃	70	30	60	30	3.5

research. Coded TF incorporated roti samples were served to the panelists randomly. A score of roti was ranked by five points hedonic scale (Rank 5 for very like to rank 1 for extremely dislike).

2.4.3. Determination of functional properties of composite flour

Functional properties of the flour blend which was used to prepare best roti (selected through the sensory evaluation) and the flour used to make the control were determined.

Swelling Capacity

Swelling capacity of flour blends was determined according to Okaka & Potter (1977). The flour sample was filled into the 10ml mark in a 100ml stoppered graduated cylinder and volume was made up to the mark of 50 ml with distilled water. Then, the cylinder was closed and mixed by inverting it. The suspension was inverted again after 2 min and allowed to stand for 30 min. The volume occupied by the sample was taken after 30 min.

Bulk Density and Tapped Density

Bulk density of samples was evaluated according to Cynthia et al., (2015) with slight modifications. Ten grams of the flour sample were measured into a 50 ml measuring cylinder and was mixed at a speed of 1500 rpm by vortex vibrator for 1 min. The ratio of mass of the flour and the volume occupied by the sample was calculated as the bulk density of the sample. The cylinder with the same contents was tapped by hand on a bench for 100 times from a height of 10 cm to determine tapped density. The ratio of the mass of the sample to volume occupied by the tapped sample was calculated as the tapped density.

Wettability

The wettability of flour samples was determined according to the method described by Vissotto et al., (2010) by measuring the time required for 1g of flour to deposit on liquid surface until it is completely submerged in 100ml of distilled water placed in 250ml beaker at 25°C.

Oil absorption and Water absorption capacity

One gram of flour sample was mixed with 100ml of distilled water and 100ml of soybean oil separately in two centrifuge tubes. Then, the samples were kept at room temperature for 30min and were centrifuged at 3000rpm for 30min. The volume of free water and soybean oil were recorded directly from the centrifuged tube. Water absorption and oil absorption were determined as percent water or oil bound per gram of flour.

2.4.4. Nutritional composition of (TF) incorporated roti and control roti

Moisture, crude fat, crude protein, crude fiber and ash contents of selected roti sample (20%TF) and 100% wheat flour roti were determined according to the methods described in AOAC (1999).

2.5. Data analysis

Each experiment was carried out with completely randomized design. Data were analyzed using both qualitative and quantitative methods. Non-parametric data were analyzed using Friedman test while parametric data were analyzed using Analysis of variance (ANOVA) using Minitab 17 statistical software package with the 0.05 significant level.

3. Results and Discussion

3.1. Quality of raw materials

The proximate composition of dehydrated coconut testa and TF are presented in Table 2.

Table 2 : Proximate composition of dehydrated coconut testa and TF

Constituent (%)	Dehydrated testa	TF
Moisture	4.04 ± 0.12	2.56 ± 0.12
Fat	58.06 ± 1.40	16.97 ± 2.44
Ash	3.39 ± 0.18	4.87 ± 0.15
Crude fiber	13.79 ± 1.93	23.51 ± 0.94
Crude protein	12.65 ± 0.25	18.02 ± 0.09
Carbohydrates	7.36 ± 0.39	32.80 ± 2.53
Free fatty acid content	1.33±0.02	NT
Peroxide value	not detected	NT

Results expressed as mean ± SD of triplicate analysis, NT; Not tested

During the drying process, the moisture content of pairing has reduced to 4.04 ± 0.12%, while it further decreased to 2.56 ± 0.12% due to oil extraction process. Moisture content is the major factor which determines the shelf-life stability of processed products (Coultate, 2009). Due to low moisture content (<3%) of the TF, it has a longer shelf life with low possibility of hydraulic rancidity and microbial growth.

Even though TF is a by-product from oil extraction process, it contains a considerable fat content (16.97± 2.44%). The residual oil in TF can be utilized as an internal fat source when it is used for food preparation. Therefore, addition of fat can be reduced. A previous study reported that the fatty acid distributions of oils extracted from TF of different cultivars of coconut contained 82.5–91.23% saturated fatty acids (SFA) and

8.76–17.5 % unsaturated fatty acid (USFA) (Marasinghe et al., 2019; Appaiah et al., 2014). According to these studies, oil present in coconut TF is low in SFA and high in USFA compared to the oil of whole coconut kernel which contains 94.6% SFA and 5.4% USFA. Therefore, coconut TF is nutritionally an important dietary ingredient.

The free fatty acid (FFA) content of oil extracted in testa is 1.33 ± 0.018%, which is generally high compared to the free fatty acid content of coconut flour obtained from white coconut kernel which is the by-product from virgin coconut oil. The coconut flour from white coconut kernel is less than 0.2% as lauric acid (FPDD, Phillipines). Therefore, the processing condition of TF should be improved to bring down the FFA content to below 0.2% value than the FFA content of crude coconut oil (max 0.4%) which is

recommended by the APCC standards 2020. Due to uncontrolled way of heating, formation of FFA can occur in brown testa even before the oil is expelled. In contrast, white coconut kernel is processed in a controlled way to reduce FFA formation. Therefore, FFA value of oil present in TF is higher than the oil present in white coconut oil.

Peroxide value indicates extent of peroxide free radicals present in the dehydrated testa. The free radicals of lipid oxidation further produce secondary products such as aldehydes, ketones and alcohols. Results show that there is no peroxide formation in TF.

Ash represents the composite form of minerals present in flour. Mineral content of foods is important because it provides minerals needed by the body. The quality of many foods depends on the concentration and type of minerals they contain. Minerals contribute to taste, appearance, texture and stability of the food. Previous studies showed that ash content of wheat flour varies from 1.17% to 2.96 % (Obert et al., 2004). The data presented in Table 2 shows that ash contents of the TF was 4.87 ± 0.15 % which is also in agreement with the guide of Food Product Development Center on Philippine (FPDD, 2019). It is comparatively higher than wheat flour (WF). Also, it is generally accepted that the ash content of flour does not affect the baking performances (Borla et al., 2004). Therefore, high ash content of TF is safe for food preparations.

According to the results of this study, protein content of coconut pairing and coconut TF were 12.65 ± 0.25 and 18.02 ± 0.09 respectively. Gunathilake et al., (2009) reported that the substitution of WF from non-wheat flour sources could increase the protein content of

bakery products. However, the presence of gluten is very essential to build the frame to trap the gases during food leavening. Therefore, substitution of wheat flour from the coconut TF, reduces the gluten content and thereby leavening characteristics of the food is affected. However, non-WF with high protein content has been used for development of food products (Nasir et al., 2020; Setia et al., 2020).

Fiber is an essential therapeutic component in daily diet to prevent from constipation, hemorrhoids, diverticulosis, coronary heart diseases some cancer and maintain body mass index. Based on the recommendation of World Health Organization (2019), adults need to take 25g of fiber daily. Hence, incorporation of fiber rich substances in the daily diet is an essential task to improve healthy life. The coconut TF has 23.51 ± 0.94 % of crude fiber. The whole WF contains low concentrations of crude fiber (0.96 ± 0.06 %) and substitution of TF from whole WF is a good option to increase fiber content of the diet.

3.2. Sensory Evaluation of TF incorporated *roti*

The results of sensory evaluation of three coconut *roti* samples which were prepared by incorporating different percentages of TF, namely, 10%, 20% and 30% (W/W) are shown in Figure 1. According to the results, all sensory attributes were significantly affected ($P < 0.05$) by the incorporation of TF into the WF for the preparation of *roti* which is an unleavened type of bread. However, there were no significant differences ($P > 0.05$) between 10% and 20% TF incorporation for all sensory attributes. However, 10% TF incorporated *roti* showed the highest acceptance for odour (63), taste (62),

texture (59) and overall acceptance (62) while better appearance is from 20% TF incorporation. With the reduction of WF proportion in the composite flour, the texture of the roti was changed due to low concentration of gluten in the composite flour as gluten imparts essential structural, rheological, and organoleptic properties of the food prepared with WF (Oniszczyk et al., 2019). Results revealed that 20% of TF can be incorporated with 80% of WF as a value addition of TF with enriching the nutritional status of roti. Gunathilake et al., (2009) reported that coconut flour prepared from dehydrated defatted desiccated coconut obtained after extraction of virgin coconut oil could successfully be incorporated at 10 and 20% with WF to prepare bread.

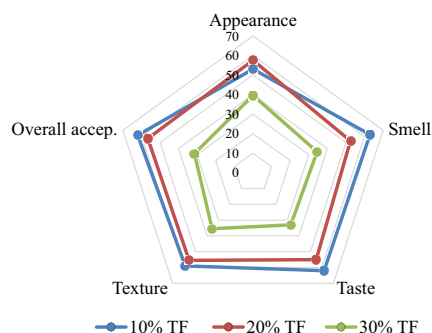


Figure 1: Sum of ranks values of sensory evaluation of TF incorporated roti

3.3. Functional Properties of Wheat Flour: Testa Flour Blends

Functional properties of selected WF blends are shown in Table 3. Results showed that there was no significant difference ($P > 0.05$) between selected flour blend (20% TF: 80% WF) with 100% WF

for swelling capacity (SC), bulk density (BD), tapped density (TD) and water absorption capacity (WAC).

Table 3 : Functional properties of 20% TF: 80% WF blend and 100% WF roti mixtures

Parameters	20% TF+ 80% WF	100%WF
SC (%)	15.67 ± 1.53^a	16.33 ± 1.53^a
BD (g/cc)	0.57 ± 0.01^a	0.52 ± 0.03^a
TD (g/cc)	0.77 ± 0.04^a	0.78 ± 0.02^a
OAC (%)	138.03 ± 4.75^a	116.46 ± 6.98^b
WAC (%)	154.60 ± 9.31^a	134.24 ± 8.33^a
WT (min)	0.34 ± 0.01^a	1.38 ± 0.08^b

The values are mean $\pm$ SD of three independent determinations. Means followed by the same letter within a row are not significantly different at $P \leq 0.05$. SC: swelling capacity, BD: bulk density, TD: tapped density, OAC: oil absorption capacity, WAC: water absorption capacity, WT: wettability

Oil absorption capacity (OAC) and wettability (WT) of the flour blend (TF 20: WF 80 blend) and wheat flour were changed significantly ($P < 0.05$). SC, BD, TD and WAC were not changed significantly due to incorporation 20% coconut TF into WF. Incorporation of TF improves the oil absorption capacity with showing repellent action for wetting. Moreover, 100% WF shows high SC (16.33%) than the 20% TF blend (15.67%). The SC of flours depends on size of particles, types of variety and type of processing methods or unit operations. BD has increased (from 0.52 to 0.57%) with the 20% of TF blending. BD depends on the particle size and initial moisture content of flours. BD of flour relates to the suitability for use in various food preparations. Low BD would be an advantage in the formulation of

complementary foods (Akapata & Akubor 1999). Suresh et al., (2015) reported that flour composite (55:15:15:15 wheat flour, rice flour, green gram flour and potato flour) with 0.820g/cc BD was suitable for biscuit preparation. High WAC of composite flour suggests that the flour can be used in the formulation of some foods such as sausage, dough, processed cheese and bakery products. However, WAC is not affected by the addition of TF. Therefore, WAC of the flour blend is suitable for similar applications as WF. The flour with high water absorption may have more hydrophilic constituents such as polysaccharides (Butt & Batool 2010). Coconut flour has high carbohydrate content and crude fibre which are having potential to attract water molecules.

Oil absorption capacity (OAC) of flour blend with 20% TF is higher (138.03 ± 4.75) than the OAC of 100% WF (116.46 ± 6.98). However, Suresh et al (2015) reported that the OAC of flour blend (Wheat, rice, green gram and potato) was increased and a blend with wheat: rice: green gram and potato at 55:15:15:15 with 156 ± 16.73 of OAC is accepted for biscuit preparation. The possible reason for the increase of the OAC of the flour blend after incorporation of TF is due to the variations of the presence of non-polar side chains present in TF, which might bind the hydrocarbon side chain of the oil. The wettability of 20% TF blend is lower ($0.34 \pm 0.01\text{min}$) than 100% WF ($1.38 \pm 0.08\text{min}$). The possible reason can be the repellent action of the TF blend which contains high oil content.

3.4. Proximate Composition Analysis

Proximate results of selected (20%) TF incorporated roti samples and control samples (100 %WF) are presented in Table 4.

Table 4 : Proximate values of 100 % WF and 20 % TF incorporated *roti*

Constituent (%)	Type of <i>roti</i>	
	100% WF	20% TF + 80% WF
Moisture	22.69 ± 0.64^a	21.48 ± 5.08^a
Fat	16.66 ± 0.72^a	15.59 ± 0.43^a
Ash	3.16 ± 0.18^a	4.32 ± 0.52^a
Crude fiber	1.80 ± 0.16^b	7.04 ± 1.15^a
Crude protein	12.43 ± 1.14^b	16.60 ± 1.02^a
Carbohydrates	40.08 ± 1.75^a	34.97 ± 4.21^a

The values are mean $\pm$ SD of three independent determinations. Means followed by the same letter within a row are not significantly different at $P \leq 0.05$. All are expressed on a wet basis TF: Testa flour incorporated roti

According to the results, there was no significant difference ($P > 0.05$) in moisture, fat, ash and carbohydrate content between TF roti and WF roti. Incorporation of 20% of TF affected significantly ($P < 0.05$) for the fiber and protein content of two treatments. The crude fiber and crude protein content of wheat flour roti has increased respectively from $1.80 \pm 0.16\%$ to $7.04 \pm 1.15\%$ and $12.43 \pm 1.14\%$ to $16.60 \pm 1.02\%$ due to the incorporation of 20% TF. Therefore, fortification with TF is more effective due to the nutritional benefits while preserving organoleptic attributes. Due to high dietary fiber and gluten free protein in TF roti, it can be introduced as a suitable traditional food for the diabetic patients.

4. Conclusion

De-fatted dehydrated coconut TF incorporation into WF for Sri Lankan traditional foods such as *roti* has considerable effect on physicochemical and sensory properties. The TF can be successfully incorporated to level of 20% to wheat flour to enhance nutritional qualities with acceptable sensory attributes.

REFERENCES

- Akapata, M.I. & Akubor, P.I. (1999). Chemical composition and selected functional properties of sweet orange (*Citrus sinensis*) seed flour. *Plant Food Human Nutrition*, 54, 353–362.
- AOAC International. (1999). Official methods of analysis of AOAC International, 17th Edition, Association of Analytical Communities, Gaithersburg.
- APCC (Asian Pacific Coconut Community) (2020). APCC standards for virgin coconut oil. Asian and Pacific Coconut Community.
- Appaiah, P., Sunil, L., Kumar, P. P. K. & Krishna, G. A. G. (2014). *Journal of American Oil Chemists' Society*, 91(6):914-926. <https://doi.org/10.1007/s11746-014-2447-9>.
- Borla, O. P., Motta, E. L., Saiz, A. I. & Fritz, R. (2004). Quality parameters and baking performance of commercial gluten flours. *LWT- Food Science and Technology*, 37, 723–729. <https://doi.org/10.1016/j.lwt.2004.02.013>
- Butt, M. S. & Batool, R. (2010). Nutritional and functional properties of some promising legumes proteins isolates. *Pakistan Journal of Nutrition*, 9(4), 373–379
- Coultate, T. P. (2009). Food: The chemistry of its components. The Royal Society of Chemistry, Cambridge, UK
- Cynthia, S. J., Bosco, J. D. & Bhol, S. (2015). Physical and structural properties of spray dried Tamarind (*Tamarind usindica* L.) pulp extract powder with encapsulating hydrocolloids, *International Journal of Food Properties*, 18(8), 1793-1800. <https://doi.org/10.1080/10942912.2014.940536>
- FPDD guide No.4, Rev 2 series (2019). Food Product Development Division, Albay Research Centre, Philippines Coconut Authority, Banao, Guinobatan, Albay, Phillipine
- Gunarathne, K. M. R. U., Marikkar, J. M. N., Mendis, B. Yalagama, C., Jayasinghe, U. L. B., Liyanage, R. & Jayaweera, S. (2022). Bioactivity of different solvent extracts of partially defatted coconut testa obtained from selected coconut cultivars. *Journal of Agricultural Sciences- Sri Lanka*, 17(1):171-184.
- Gunathilake, K. D. P. P., Yalagama, C. & Kumara, A. A. N. (2009). Use of coconut flour as a source of protein and dietary fibre in wheat bread. *Asian Journal of Food and Agro-Industry*, 2(3), 382-391

- Jayatissa, R. L. N., Wickramasinghe, W. D. & Piyasena, C. (2014). Food consumption patterns in Sri Lanka. Hector Kobbekaduwa Agrarian Research and Training Institute.
- Marasinghe, S. S. K., Nazrim, J. M., Marikkar, Yalgama, C., Wimalasiri, S., Seneviratne, G., Weerasooriya, R. & Liyanage, R. (2019). Comparison of inter-varietal differences in chemical composition and nutritional properties of coconut testa flour. *Journal of the National Science Foundation of Sri Lanka*, 47(3), 349-356
- Nasir, S., Allai, F. M., Gani, M., Ganaie, S., Gul, K., Jabeen, A. & Majeed, D. (2020). Physical, textural, rheological and sensory characteristics of amaranath-based wheat flour bread. *International Journal of Food Science*, 2020, ID 8874872 <http://doi.org/10.1155/2020/8874872>.
- Obert, J. C., Ridley, W. P., Schneider, R. W., Riordan, S. G., Nemeth, M. A., Trujillo, W. A., Breeze, M. L., Sorbet, R. & Astwood, J. D. (2004). The composition of grain and forage from glyphosate tolerant wheat MON 71800 is equivalent to that of conventional wheat (*Triticum aestivum* L.). *Journal of Agricultural Food Chemistry*, 52, 1375-1384.
- Okaka J. & Potter N. N. (1977). Functional and storage properties of cowpea-wheat flour blends in bread making. *Journal of Food Science*, 42, 828-833.
- Oniszcuk, A., Widelska, G., Wójtowicz, A., Oniszcuk, T., Wojtunik-Kulesza, K., Dib, A. & Matwijczuk, A. (2019). Content of phenolic compounds and antioxidant activity of new gluten-free pasta with the addition of chestnut flour. *Molecules*, 24, 2623.
- Perera, M. P. M. S. H., Sivakanesan, R., Abeysekara, D. T. D. J. & Sarananda, K. H. (2015). Glycemic response of traditional breakfast foods in adults with type 2 diabetes mellitus. *Tropical Agricultural Research*, 27(1), 39-48.
- Ranasinghe, P., Jayawardena, R. & Katulanda, P. (2015). The facts, figures, and reality of the diabetes epidemic in Sri Lanka: a systematic review. *International Journal of Diabetes in Developing Countries*, 35, 501-513. <http://doi.org/10.1007/s13410-015-0326-z>.
- Rushdha, S. A. F., Uipathakumbura, B. S. K., Yalgama, C., Hewapathirana, D. T. & Marikkar, J. M. N. (2022). Evaluation of staple foods supplemented with defatted coconut testa flour. (Accepted for publication). *Cord*, 38.
- Sami, W., Ansari, T., Butt, N. S. & Hamid, M. R. A. (2017). Effect of diet on type 2 diabetes mellitus: A review. *International Journal of Health Sciences*, 11(2), 65-71.
- Setia, R., Dai, Z., Nickerson, M. T., Sopiwnyk, E., Malcolmson, L. & Ai. Y. (2020). Properties and bread making performance of wheat flour composited with germinated pulse flour. *Cereal Chemistry*, 97, 459-471.
- Sri Lanka Standards Specification for coconut oil (2002). SLS 32:2002. Sri Lanka Standards Institution, 17, Victoria Place, Elvitigala Mawatha, Colombo 8, Sri Lanka.

Suresh, C., Samsher, S. & Durvesh, K. (2015). Functional properties of composite flour & sensorial attributes of composite flour biscuits. *Journal of Food Science & technology*. 52(6),3681-3688.

Thathvasuthan, A., Chandrasekara, A. & Wijesinghe, D. G. N. G. (2007). The effect of type of cereal on the glycaemic response of two traditional Sri Lankan food items. *Tropical Agricultural Research*, 19 (January), 101–109.

Vissotto, F. Z., Jorge, L. C., Makita, G. T., Rodrigues, M. I. & Menegalli, F. C. (2010). Influence of the process parameters and sugar granulometry on cocoa beverage powder steam agglomeration. *Journal of Food Engineering*, 97, 283–291.

World Health organization (2019). Retrieved on 12/03/2022 from [Zmescience.com/Science/who-report-fibres-14012019](https://www.zmescience.com/Science/who-report-fibres-14012019).

RESEARCH ARTICLE

Fighting Drought: A Mini Review with Special Reference to Tropical Palms

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ABSTRACT

Drought stress is the most important environmental stress which affects negatively on plant growth and productivity. With the climate change and the scarcity of water, drought conditions have become frequent and prominent, affecting the agricultural production worldwide. Therefore, breeding crops for drought tolerance has gained significant attention at present. To cope with drought, plants possess different mechanisms and understanding their biological significance would enable the utilization of these mechanisms in crop improvement for drought tolerance. Therefore, in the current review an effort has been made to compile the information related to drought tolerance mechanisms and drought resistant/tolerant crop improvement with special reference to tropical palm tree species.

Keywords: Drought, Drought tolerance, Palms

INTRODUCTION

Among the negative impacts of severe environment on plants, drought can be considered as one of the most significant environmental stresses that affects the plant growth and performance. Drought stress occurs as a result of an imbalance between the evapotranspiration flux and water intake from the soil (Lipiec et al., 2013) where the soil and atmospheric humidity becomes lower than the ambient air temperature. The adverse effects of drought stress involve mainly the reduction of crop productivity and yield. For instance, 21% and 40% of yield reduction has been observed in wheat and maize crops respectively due to drought stress caused by approximately a 40% reduction of water (Daryanto et al., 2016). Currently, about 45% of the world's total agricultural land has been affected by drought (Dos Reis et al., 2012). The effect of drought has become more severe with the climate change. Intergovernmental Panel on Climate Change (IPCC) reports that the air and ocean temperatures have risen, and the concentrations of greenhouse gases have increased over time (Edenhofer, 2015). Both these factors affect the plant growth and crop productivity (Stocker et al., 2013) as they result in elevated temperatures and infrequent and erratic precipitations and thereby contributing to the development of drought-prone areas (Zhao et al., 2017). Therefore, with the increment of global warming and rising of population, it is crucial to take measures for the development of drought tolerance in crops, in order to meet the demand. In this review, the efforts that have been made so far to improve crop productivity under water-limiting conditions, and to develop drought tolerant crops have been discussed with special reference to tropical palm trees.

Drought tolerance in plants

Drought tolerance is defined as the ability of a plant to survive, grow, and reproduce satisfactorily with limited water supply (Turner, 1979). Crop plants should have both the ability to survive under drought and the ability to produce a harvestable yield under adverse conditions. In order to cope with environmental stresses, plants possess different strategies and among them, drought escape, dehydration avoidance and dehydration tolerance are important in facing and responding to drought (Levitt, 1980). Simply, drought escape strategies include the completion of the full life cycle of the plant prior to approaching a drought event. Dehydration avoidance strategies include the morpho-physiological features that enable the plant to maintain hydration whereas dehydration tolerance strategies include the features that allow the plant to maintain proper functionality in severe dehydrated state. However, these three broad strategies include a cascade of physiological, biochemical and molecular events and they will be discussed in detail.

Drought Escape

As mentioned earlier drought escape can be described as the ability of a plant species to complete its life cycle before the commencement of drought. Plants which possess this strategy are usually ephemerals and they can regulate the vegetative and reproductive growth depending on the availability of water, thereby avoiding the experiencing of the drought stress. Two main mechanisms are identified in association to this strategy namely the rapid phenological development and development plasticity (Basu et al., 2016). Rapid phenological

development includes the rapid plant growth with early flowering and production of minimal number of seeds before the depletion of the soil water (Abobatta, 2019). However, such plants do not possess with any special morphological, physiological, or biochemical adaptations for drought (Basu et al., 2016). Plants with mechanisms of developmental plasticity show retarded growth during the dry season, with very few flowers and seeds, but in wet seasons they grow indeterminately, producing a large number of seeds.

Dehydration avoidance

When considering the morphological adjustments of plants to cope with heat stress or drought, the root system lies in front. The root of the plant can be considered as the key organ providing mechanical support and it facilitates the absorption and transportation of water, oxygen, nutrients, and hormones to the aboveground tissues. As it has an important role in the abiotic stress tolerance, crop performance and yield production in response to environmental variations, the root modifies its architecture to a level in which it increases the water uptake efficiency. In response to drought stress, many plants prefer to increase the primary root elongation while suppressing the lateral root branching (Oosten et al., 2016). Studies have identified several genetic components contributing to root architecture that has the potential to limit crop loss due to adverse environmental conditions. Moreover, plants re-model the root system during stresses by activating programmed cell death in the root apical meristem and thereby removing the apical root dominance (Oosten et al., 2016). Drought tolerance is a complex composite resulting from the interaction of root and shoot traits (Lu et al., 2007). The size and activity of the root system determines the rate at which the

shoot system can produce photosynthates. Therefore, despite of the less well studied nature of roots due to lack of ready visibility and issues in replicating the conditions in which they grow, there is a higher potential in exploitation and manipulation these root traits to optimize crop growth and adaptation to water stress. (Den Herder *et al.*, 2010).

Physiological adjustments that maintain turgor pressure are important under changing environmental conditions. Water transportation in roots is affected by various components such as root anatomy, water availability and salts in the soil (Boursiac *et al.*, 2005). All these factors are influenced by the activity of aquaporins, which are integral membrane proteins that function as channels to transfer selected small solutes and water (Vandeleur *et al.*, 2014).

Stomata is another key player in dehydration avoidance as they control and regulate the evapo-transpiration of water to the external environment (Pirasteh-Anosheh et al., 2016). Stomata usually occur mostly on the lower surface of the leaf and new stomata are generated continually during leaf growth. Stomatal closure is a well-known first responsive reaction of plants to drought stress in many crops (Nemeskeri et al., 2015). Stomatal closures are more closely related to soil moisture content compared to leaf water status. Stomatal responses to drought stress are mainly controlled by chemical signals such as ABA that are produced in dehydrating roots. According to Taiz and Zeiger (2002), stomatal closure can be observed under severe drought conditions along with other physiological processes such as inhibition of cell division, inhibition of wall and protein synthesis, accumulation of solutes and inhibition of photosynthesis. However, stomata closure does not always depend on the precipitation

or drought stress signals arising from leaves. In fact, stomata closure is directly sensitive to soil dehydration, even before any significant reduction in leaf mesophyll turgor pressure occurs (Mahajan and Tuteja, 2005).

Many studies have indicated that drought stress results in increasing stomatal density in sorghum, wheat and in some other plants (Yang and Wang, 2001; Xu and Zhou, 2008). Stomatal morphological traits such as size, density, and distribution may affect gas exchange quite remarkably and their relationships with key environmental factors such as water status (Xu and Zhou, 2008). Moreover, some studies have shown that drought stress decreased stomatal size, which indicated these changes in stomatal morphology could increase the plant adaptation to drought stress conditions (Martinez et al., 2007). Yadollahi et al. (2011) revealed that leaf stomatal cells of an Almond had strong mechanisms to keep the leaves turgid and active, even during drought periods. The lack of relationship between stomatal density and drought stress in some plants showed that they possibly use another mechanism for drought resistance such as stomatal conductance.

Dehydration tolerance

Exposure to drought leads to cellular dehydration where water removes from the cytoplasm resulting in decreasing of cytosolic and vacuolar volumes (Forni et al., 2017). In response to this, plants increase the production of specific sets of primary and secondary metabolites that act as osmo-protectants, osmolytes, antioxidants, and stress signals which are specifically involved in stress tolerance (Fang and Xiong, 2015).

Osmo-protectants involve in preserving the cells and cellular components from the

detrimental changes caused by dehydration, without interfering with the normal metabolic processes at the cellular level and thereby restoring homeostasis. These solutes include amines (polyamines and glycine betaine), amino acids (proline), soluble sugars (glucose, sucrose, trehalose), and polyols (mannitol, sorbitol and inositol) (Szabados and Savouré, 2010). The accumulation of these compounds aid in protection of protein structures and stabilization of cellular membranes as cells dehydrate, maintenance of cell turgor and thereby providing abiotic stress tolerance (Zlatev and Lidon, 2012).

However, some crops have low levels of these compounds, and hence such plants manipulate the genes involved in osmo-protectant biosynthesis pathways and it is considered as one of the strategies to improve stress tolerance in plants (Reguera et al., 2012).

Polyamines (PAs) are small aliphatic nitrogen compounds that are ubiquitous in all organisms. In plants, PAs act as regulatory molecules involved in vital cellular processes such as embryogenesis, floral development, and pollen tube growth (Tiburcio et al., 2014). A significant accumulation of the three most common PAs, namely; Putrescine (Put), Spermidine (Spd), and Spermine (Spm) could be identified during the biotic and abiotic stress conditions (Wen and Moriguchi, 2015).

Glycine Betaine (GB) is a quaternary ammonium derivative of glycine and is one of the major osmolytes involved in protection of the cell membrane. They are involved in encouraging water influx into cells and maintain the intracellular osmotic equilibrium while regulating the signal transduction cascades (Ranganayakulu et al., 2013). In many plant species these GBs are accumulated

in chloroplasts and other plastids in response to various abiotic stresses such as drought.

Proline, a versatile amino acid that is essential both as a component of the protein and as a free amino acid contributing to osmotic adjustment in response to different abiotic stresses (Zandalinas *et al.*, 2017). Proline facilitates water uptake and reduces the accumulation of Na^+ and Cl^- (Ashraf and Foolad, 2007) and avoid dehydration. Therefore, accumulation of proline under stress conditions in many plant species has been correlated with stress tolerance.

Other than the above-mentioned compounds, plant hormones also play major roles in abiotic stress tolerance. Absciscic acid (ABA) is one of the most important hormones involved in abiotic stress tolerance such as drought, cold, heat and wounding (Zhang *et al.*, 2006). ABA also is considered as a major chemical involved in root-to-shoot stress signal transduction (Schachtman and Goodger, 2008), inducing inhibition of leaf expansion and stomatal closure. Osmotic stress also results in the synthesis or catabolism of several other growth regulators, including Auxin, Cytokinins, Ethylene, Gibberellins, Brassinosteroids and Jasmonic acid which are involved in the regulation of physiological processes through their action as signal molecules in signaling networks (Nakashima and Yamaguchi-Shinozaki, 2013).

In addition to the above strategies, plants also enhance intracellular Ca^{2+} levels with the support of several signal molecules such as Inositol Trisphosphate, Inositol Hexaphosphate and Diacylglycerol (Hirayama and Shinozaki, 2010) during stress. Calcium binding proteins,

functioning as Ca^{2+} sensors, perceive these elevated Ca^{2+} levels (Kudla *et al.*, 2010) and it will lead to the activation of calcium dependent protein kinases. The activated kinases or phosphatases can phosphorylate or dephosphorylate specific transcription factors (TFs), thus regulating the expression levels of stress-responsive genes (Reddy *et al.*, 2011). TFs play important roles in stress tolerance. Many abiotic stress-related genes and TFs have been isolated from different plant species mainly in wheat and maize and overexpressed in transgenic plants to improve stress tolerance. The stress-inducible TFs include members of the DREB, ERF, WRKY, MYB, bHLH, bZIP, DOF, and NAC families (Mittal *et al.*, 2017).

Another important cellular mechanism conferring drought tolerance is the regulation of Reactive Oxygen Species (ROS) (Hirayama and Shinozaki, 2010). ROS are chemically reactive molecules containing oxygen and they are formed in the metabolism of oxygen and have important roles in cell signaling and homeostasis. Increased levels of ROS have been reported during biotic and/or abiotic stresses, such as pathogen attack, wounding, UV irradiation, high light, drought, salinity, and chilling (Sharma *et al.*, 2012). Accumulation of ROS causes oxidative stress, which in turn results in oxidative damage to proteins, DNA and lipids (Gill and Tuteja, 2010). Acclimation of plants to drought and salinity is often associated with increased levels of ROS, such as superoxide anion ($\text{O}_2^{\cdot -}$), hydrogen peroxide (H_2O_2), hydroxyl radical ($\text{HO}\cdot$), and singlet oxygen (O_2), which are toxic to the cells (Grover *et al.*, 2011). During non-stress conditions, ROS are efficiently eliminated by non-enzymatic and enzymatic antioxidants. However, during drought and salt stress, the

production of ROS exceeds the capacity of the anti-oxidative systems to remove them, causing oxidative stress (Vanderauwera et al., 2012). In these conditions, the elimination of ROS is a key response to tolerate drought stress. This is mainly achieved by antioxidant compounds such as ascorbic acid, glutathione, thioredoxin, and by oxyreductant enzymes such as glutathione peroxidase, superoxide dismutase, and catalase. It has been established that 1–2 % of oxygen absorbed by plants is used to produce ROS in plants (Bhattacharjee, 2005). Over accumulation of ROS from abiotic stress contributes to major losses of crop productivity and is an important economic problem for cultivated plants worldwide (Gill and Tuteja, 2010).

The above discussed different strategies used by plants as drought tolerance mechanisms are summarized in Figure 1.

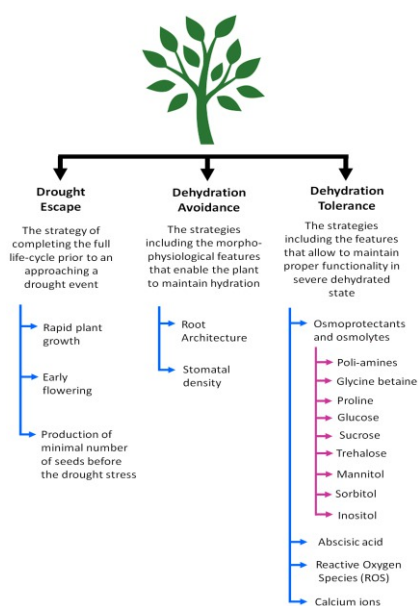


Figure 1: Different mechanisms used by plants to cope with drought stress (Only the major strategies are illustrated.)

Improving drought tolerance in crops

With the scarcity of water and frequent droughts due to climate change, the worldwide agricultural ecosystems are facing the challenge of combatting drought. Therefore, breeding for water saving and drought tolerant crops is a priority in the current world for ensuring world food security. A significant progress has been achieved in this aspect during the last decade (Luo et al., 2019) with the advancement of technology and novel findings.

Breeding strategies include several methodologies such as breeding and selection in the field, Quantitative Trait Loci (QTL) mapping or Genome Wide Association Studies (GWAS), comparative transcriptome analysis, precise high-throughput phenotyping, and characterization of drought-resistance genes, to induce drought-resistance in different crops.

Understanding the genetic basis of drought tolerance in plants is vital in developing superior genotypes resistant to drought. Generally, drought tolerance is a quantitative trait (Fleury *et al.*, – 2010) controlled by multiple genes. Therefore, the underpinning mechanisms of drought tolerance are more complex than the traits of biotic stresses that are generally characterized by monogenic resistance. However, studies have shown that single genes also are important in drought tolerance in plants (Morgan, 2000) and these genes include the genes controlling flowering time, plant height, and osmotic adjustment which might have important roles in adaptation to drought-prone environments (Forster *et al.*, 2004).

At present, majority of the data available on drought tolerance are based on segregation mapping and QTL analysis. Identification of drought-resistance QTLs is essential to provide valuable targets in crop breeding. Drought related QTL information for majority of the plant species is available in open source databases like GRAMENE (<http://www.gramene.org>) and Phytozome (<https://phytozome.jgi.doe.gov/pz/portal.html>) which can be utilized freely for further research. When considering the drought related QTLs for several crops they coincide with the other specific traits such as yield and root traits (Verma et al., 2004) which can be used in improving drought tolerance in crops using marker assisted selection. For instance, Lanceras et al. (2004), showed that favorable alleles for yield components were located in a region of rice chromosome 1 where QTLs for many drought-related traits like root dry weight, relative water content, leaf rolling and leaf drying which have been previously identified. However, due to the polygenic nature of drought tolerance and the interaction with the environment, a genetic linkage between a specific DNA marker and a target locus allele established by QTL studies can be broken. As a result, Marker Assisted Selection for drought tolerant QTLs has been extremely difficult (Francia et al., 2005). Cloning of DNA sequences underlying QTLs is considered to be a key step towards the application of molecular techniques in breeding for drought. Even though most plant QTLs have been cloned so far, no QTL for drought tolerance has been cloned in crop species (Cattivelli et al., 2008). However, ERECTA gene, a sequence beyond a QTL for transpiration efficiency has been cloned in Arabidopsis (Masle et al., 2005).

Generation of molecular/function maps or molecular linkage maps based on candidate genes is one of the means in determining the functional markers in plants with larger genomes (Causse et al., 2004). As this candidate gene strategy can be utilized to bridge the gap between quantitative genetic and molecular genetic approaches to study complex traits, it has been applied to identify the putative genes involved in drought tolerance in barley and rice (Zheng et al., 2003; Tondelli et al., 2006). With the benefit of the conserved biochemical pathways which involves in stress tolerance in plants, the genetic information in model plants can be utilized to exploit the genome synteny in crops. Several studies have identified the regulatory components of the drought response in soybean, wheat and barley using this approach (Li et al., 2005; Shen et al., 2003; Mare' et al., 2004).

At present, the novel approaches such as GWAS have broaden up the horizons in stress tolerance breeding. For example, Li et al. (2018) have identified four genomic regions containing Single Nucleotide Polymorphisms (SNPs) significantly associated with several different traits in chickpea under drought stress. This result has indicated the pleiotropic effects of drought-resistance associated QTLs. Moreover, Gudys et al. (2018) have identified 11 candidate QTLs of physiological and biochemical traits associated with drought-tolerance in Barley on a high-density function map. Furthermore, they have prioritized 143 candidate genes by their potential involvements in certain biological processes based on Gene Ontology annotation. Cui et al. (2018) has provided a new method to identify QTLs for drought-tolerance in rice by comparing the allele

frequency between drought-resistant introgression lines and random populations and identified 13 major QTLs of drought-tolerance in rice using the joint segregation distortion method. The most exciting result of this study was the detection of the large-effect QTLs locate upstream of the genetic networks as putative regulators, which could contribute significantly to drought-tolerance in rice. In addition, they also suggested the designed QTL pyramiding strategy that is feasible for improving drought-tolerance in rice. Even though there are hundreds of QTLs for the diverse traits related to drought resistance have been mapped, only a small portion of them can be repeatedly detected in different environments and populations and a few have been verified and cloned so far (Hu and Xiong, 2014).

Selection and breeding in the field have been used as the fundamental strategy in different crop species to build up drought tolerance in the upcoming progenies (Sallam et al., 2019). A crucial aspect in all field studies dedicated to drought tolerance is the assessment of the degree of drought tolerance of different genotypes (Avramova et al., 2016). However, in many studies the identification of tolerant and susceptible cultivars is based on a few physiological measures related to drought response. With the difficulty in identifying a physiological parameter as a reliable indicator of yield in dry conditions, it has been suggested that the yield performance over a range of environments could be used as the main indicator for drought tolerance (Voltas et al., 2005). Furthermore, a number of regression techniques of yield against environmental indices as independent variables were developed to evaluate genotype adaptability. Several indices were

proposed to describe yield performance of a given genotype under stress and non-stress conditions or in comparison with the average yield or the yield of a superior genotype (Finlay and Wilkinson, 1963).

Improving drought tolerance in tropical palms

When considering the attempts made so far to improve drought tolerance in perennial tree crops, it is interesting to look at the improvement of the palm species such as oil palm (*Elaeis guineensis* Jacq.), date palm (*Phoenix dactylifera* L.) and coconut (*Cocos nucifera* L.). Oil palm and coconut are the two most important plantation crops grown in the wet tropics. Even though oil palm is considered as a drought tolerant crop, negative impact of drought on the yield can be identified. Several attempts have been made to identify drought tolerant genotypes of oil palm via selection and progeny testing in the field as well as in controlled environments (Ubara et al., 2017). Physiological parameters such as the root biomass, potential root extraction ratio, rate of stomatal conductance and photosynthesis were used in these studies (Murugesan et al., 2017). Also, studies have indicated that the tolerant oil palm genotypes consisted of higher total root length, total root surface area and potential root water extraction ratio than the susceptible genotypes based on half-distances between roots and the distance of water migration from soil to root (Nodichao et al., 2011).

Furthermore, interspecific hybrids like (*E. guineensis* × *E. oleifera*) have been produced and they have shown positive responses for drought tolerance (Barcelos et al., 2005).

Application of Silicon (Si) increases oil palm resistance to drought (Sacala, 2009), by aiding

the oil palm plants to maintain water contents in the tissue, and increase photosynthetic activity (Bharwana et al., 2013), to support the establishment of leaves, to maintain the structure of xylem vessels under conditions of rapid transpiration rate, to improve the balance of nutrients, to reduce mineral toxicity and to increase the mechanical strength of plant tissues (Sacala, 2009). Other benefits of application of Si are greater water use efficiency by reducing the rate of water loss, elimination of cuticular transpiration and increase in the CO₂ assimilation rate and stomatal conductance (Sacala, 2009).

From recent studies conducted using genomic and transcriptomic approaches, genes, pathways and networks responding to drought have been identified in oil palm (Wang et al., 2020). This study on root transcriptomic response of oil palm seedlings to drought stress has revealed over thousand differentially expressed genes. These genes were involved in different molecular processes implying the complex nature of the response to drought. The differently expressed genes have been identified under different functional groups including cell wall biogenesis and functions, phenylpropanoid biosynthesis and metabolisms, ion transport and homeostasis and cellular ketone metabolic process, as well as small molecule biosynthetic processes.

Date palm (*Phoenix dactylifera* L.) is another important tree crop contributing to the food security in arid regions of the world particularly in the Middle East and North Africa (Hazzouri et al., 2020). Irrespective of the drought tolerant quality gifted by nature, studies have been conducted in several research areas to explore and improve the drought tolerance in date palm. According to

Xiao et al. (2019), date palm develops a tuber-like anatomical structure known as the cotyledonary petiole and it helps to protect the developing embryo in the soil. Furthermore, this study reported that date palms maintains suberized and lignified xylem, phloem and bundle cells in roots and also produce pneumatophores accounting for the drought and salinity tolerance.

Yaish (2015) studied the possibility of using proline as a biochemical marker in selecting for drought tolerance and found that date palm seedlings accumulate proline not only in response to drought but also as a response to salinity, extreme temperatures and abscisic acid treatments. As a result, it was concluded that the proline would be an unsuitable marker to use in traditional breeding programmes. Even though proline production is a common response for a variety of stresses, there is a potential for utilizing proline production as a possible marker in date palm breeding programs aiming to improve abiotic stress tolerance.

With the improvement of cutting-edge technologies, several omics studies also have been conducted on the drought tolerance of date palm. For example, genes involved in salt and drought tolerance in *P. dactylifera* have been identified by a proteomics study (Rabey et al., 2016). The group have identified 47 genes in total of which 17 are responsive for drought stress. Furthermore, Safronov et al. (2017) have conducted a transcriptomic and metabolomic profiling to characterize the response to heat and drought stress in date palm. The results have shown similar effects on both the stresses including the up regulation of soluble carbohydrates and increased antioxidant activity in the cytosol, chloroplasts, and peroxisomes.

Moreover, differentially expressed genes involved in circadian and diurnal rhythm in response to combined heat and drought were reported in the study implying a novel stress-avoidance strategy in date palm.

When considering coconut, it is mainly a rain-fed crop and faces summer drought spells each year in addition to the frequent drought years. Due to its perennial nature, coconuts have a long duration of about 37 months from the initiation of inflorescence primordia to nut maturity which contains a longer pre-fertilization period (About 26-27 months) and a shorter post fertilization period (11 months) (Perera et al., 2010; Ranasinghe, 2013). Therefore, any drought spell experienced at any of these critical stages affect the nut yield and the performance of the palm for about three consecutive years (Naresh Kumar, 2002). As it takes about 4 years for a coconut palm to recover in the worst affected conditions under a stress, improving the drought tolerance has been the need of the hour.

For the improvement of coconut for drought tolerance, the fundamental and promising approach which have been used is selection and progeny testing. Different coconut germplasm has been screened so far based on the essential anatomical and physiological traits such as the cell size and number, the size of the sub stomatal cavity, stomatal frequency, thickness of the epicuticular wax on leaflets, photosynthetic rate and water use efficiency (Naresh Kumar et al., 2000). Therefore, such desirable traits which reflects overall palm–water relationships could be used in identifying coconut cultivars with different levels of drought tolerance (Rajagopal et al., 2000). Based on drought tolerance indexes the germplasm also have

been screened. For example, in Sri Lanka, the two cultivars San Raman tall and Ambakelle special have been identified as drought tolerant (Wikramartne, 1987). Ambakelle special is a cultivar that is resulted from a selection of the tall coconut palms providing stable yield even under adverse climatic conditions over 16 years from the tall coconut population in the Isolated seed garden at Ambakelle, Sri Lanka. Currently, three generations of Ambakelle special cultivar have been produced by paired crossing between the selected palms and they are in further evaluation at multi-locational sites for drought tolerance (Samarasinghe et al., 2018a; Samarasinghe et al., 2021).

In Cote d'Ivoire, PB-121 was identified as drought tolerant while West African Tall was identified as moderately tolerant to drought. According to the studies conducted at the Central Plantation Crops Research Institute (CPCRI), India have indicated that the Tall palms and the hybrids with the mother parent being Talls had shown a higher degree of drought tolerance than the dwarf palms and the hybrids with the mother parent being dwarf (Rajagopal and Kumar, 2005). With this background, in Sri Lanka, several attempts were taken to develop drought tolerant coconut cultivars through inter and intra varietal hybridization. For example, the CRISL98 (TSR) produced in 1998 by crossing of Sri Lankan Tall x San Raman Tall (Philippine cultivar) was initially recommended as a cultivar suitable to grow in the dry zone with drought tolerant traits. However, these inter and intra varietal hybrids developed with unselected Sri Lankan tall germplasm failed to develop cultivars withstanding drought and temperature stresses (Samarasinghe et al., 2018b; Waidyarathne

and Perera, 2020; Samarasinghe et al., 2021) and the recommendation of CRISL98 has been revised as to grow them in wet and intermediate zones only due to the poor performance in the dry zone. Therefore, attempts have been currently made to use the putative drought tolerant second generation Ambakelle special palms as the maternal parent in developing new coconut cultivar CRISL2020.

Use of molecular techniques and novel omics approaches for improving the drought tolerance in coconut has been attempted at CPCRI (Rajagopal and Kumar, 2005) and some of the putative stress responsive genes to water deficit have been uncovered (Ramesh et al., 2020). Furthermore, they revealed differential molecular response of genotypes to water stress.

Even though the omics outcomes of drought and other abiotic stresses are nonspecific (Yaish, 2015) and there are only a few records of potential molecular markers for improving the drought tolerance, several attempts are in progress to uncover the putative genes/metabolic pathways accounting for drought tolerance in coconut and other palm species using integrated approaches and to utilize them accordingly in improving the drought tolerance in coconut.

Conclusion

Drought tolerance is a quite complex phenomenon which is controlled by a combination of anatomical, physiological and biochemical parameters. At different developmental stages, plants adopt different mechanisms to confer drought resistance. Even though it is difficult to evaluate drought resistance of a given plant species, several attempts have been made and are in progress to uncover the secrets behind drought stress

tolerance in plants. However, in this genomic era it is important to have an integrated approach of omics tools to unwrap the linkage of traits, metabolic networks and protein interactions associated with drought stress tolerance in plants and to utilize them accordingly in improving the drought stress tolerance in plants.

REFERENCES

- Abobatta W. F. (2019). Drought adaptive mechanisms of plants – a review. *Advances in Agriculture and Environmental Science*. 2(1): 42–45. DOI: 10.30881/aaea.00021
- Al-Amin, A. Q., Leal, W., de la Trinxeria, J. M., Jaafar, A. H. and Ghani, Z. A. (2011). Assessing the impacts of climate change in the Malaysian agriculture sector and its influences in investment decision. *Middle-East Journal of Scientific Research*, 7(2), 225-234.
- Ashraf, M. F. M. R. and Foolad, M. R. (2007). Roles of glycine betaine and proline in improving plant abiotic stress resistance. *Environmental and experimental botany*, 59(2), 206-216.
- Avramova, V., Nagel, K. A., AbdElgawad, H., Bustos, D., DuPlessis, M., Fiorani, F. and Beemster, G. T. (2016). Screening for drought tolerance of maize hybrids by multi-scale analysis of root and shoot traits at the seedling stage. *Journal of Experimental Botany*, 67(8), 2453-2466.

- Barcelos, E., Cunha, N.R.W., & Nouy, B. (2005). Genetic resources of oil palm (*Elaeis guineensis* Jacq. and *Elaeis oleifera* (Kunth) Cortes) available at EMBRAPA and its use. In: BORÉM, A and VIANA, GV Plant Breeding, Viçosa, UFV. 525 p
- Basu, S., Ramegowda, V., Kumar, A. and Pereira, A. (2016). Plant adaptation to drought stress. F1000Research, 5, F1000 Faculty Rev - 1554. <https://doi.org/10.12688/f1000research.7678.1>
- Bharwana, S. A., Ali, S., Farooq, M. A., Iqbal, N., Abbas, F. and Ahmad, M. S. A. (2013). Alleviation of lead toxicity by silicon is related to elevated photosynthesis, antioxidant enzymes suppressed lead uptake and oxidative stress in cotton. *Journal of Bioremediation and Biodegradation* 4(4), 187.
- Bhattacharjee, S. (2005). Reactive oxygen species and oxidative burst: roles in stress, senescence and signal transduction in plants. *Current Science*, 89, 1113–1121.
- Boursiac, Y., Chen, S., Luu, D. T., Sorieul, M., van den Dries, N. and Maurel, C. (2005). Early effects of salinity on water transport in *Arabidopsis* roots. Molecular and cellular features of aquaporin expression. *Plant physiology*, 139(2), 790–805.
- Cattivelli, L., Rizza, F., Badeck, F. W., Mazzucotelli, E., Mastrangelo, A. M., Francia, E. and Stanca, A. M. (2008). Drought tolerance improvement in crop plants: an integrated view from breeding to genomics. *Field crops research*, 105(1-2), 1-14.
- Causse, M., Duffe, P., Gomez, M. C., Buret, M., Damidaux, R. *et al.* (2004). A genetic map of candidate genes and QTLs involved in tomato fruit size and composition. *Journal of Experimental Botany*, 55(403), 1671–1685.
- Cui, Y., Zhang, W., Lin, X., Xu, S., Xu, J. and Li, Z. (2018). Simultaneous improvement and genetic dissection of drought tolerance using selected breeding populations of rice. *Frontiers in plant science*, 9, 320.
- Daryanto, S., Wang, L., and Jacinthe, P. A. (2016). Global synthesis of drought effects on maize and wheat production. *PloS one*, 11(5), e0156362.
- Den Herder, G.D., Isterdael, G.V., Beeckman, T. and Smet I.D. (2010) The roots of a new green revolution. *Trends Plant Sci* 15:600–607. doi:[10.1016/j.tplants.2010.08.009](https://doi.org/10.1016/j.tplants.2010.08.009)
- Dos Reis, S. P., Lima, A. M. and de Souza, C. R. (2012). Recent molecular advances on downstream plant responses to abiotic stress. *International journal of molecular sciences*, 13(7), 8628–8647. <https://doi.org/10.3390/ijms13078628>
- Eberhart, S. T., & Russell, W. A. (1966). Stability parameters for comparing varieties 1. *Crop science*, 6(1), 36–40.
- Edenhofer, O. (2015). *Climate change 2014: mitigation of climate change-IPCC* (Vol. 3). Cambridge University Press.
- Fang, Y. and Xiong, L. (2015). General mechanisms of drought response and their application in drought resistance improvement in plants. *Cellular and molecular life sciences*, 72(4), 673–689.

- Finlay, K. W. and Wilkinson, G. N. (1963). The analysis of adaptation in a plant-breeding programme. *Australian journal of agricultural research*, 14(6), 742-754.
- Fleury, D., Jefferies, S., Kuchel, H. and Langridge, P. (2010). Genetic and genomic tools to improve drought tolerance in wheat. *Journal of experimental botany*, 61(12), 3211-3222.
- Forni, C., Duca, D. and Glick, B. (2017). Mechanisms of plant response to salt and drought stress and their alteration by rhizobacteria. *Plant and Soil*, 410(1/2), 335-356
- Forster, B. P., Ellis, R. P., Moir, J., Talame, V., Sanguineti, M. C. *et al.* (2004). Genotype and phenotype associations with drought tolerance in barley tested in North Africa. *Annals of applied Biology*, 144(2), 157-168.
- Francia, E., Tacconi, G., Crosatti, C., Barabaschi, D., Bulgarelli, D., Dall'Aglio, E. and Valè, G. (2005). Marker assisted selection in crop plants. *Plant Cell, Tissue and Organ Culture*, 82(3), 317-342.
- Gill, S. S. and Tuteja, N. (2010). Reactive oxygen species and antioxidant machinery in abiotic stress tolerance in crop plants. *Plant physiology and biochemistry*, 48(12), 909-930.
- Grover, A., Singh, A. and Blumwald, E. (2011). Transgenic strategies toward the development of salt-tolerant plants. *Agricultural salinity assessment and management*, 2nd edn, pp. 235-274.
- Gudys, K., Guzy-Wrobelska, J., Janiak, A., Dziurka, M. A., Ostrowska, A., Hura, K. and Szarejko, I. (2018). Prioritization of candidate genes in QTL regions for physiological and biochemical traits underlying drought response in barley (*Hordeum vulgare* L.). *Frontiers in plant science*, 9, 769.
- Hazzouri, K.M., Flowers, J.M., Nelson, D., Lemansour, A., Masmoudi, K. and Amiri, K.M.A. (2020). Prospects for the Study and Improvement of Abiotic Stress Tolerance in Date Palms in the Post-genomics Era. *Frontiers in plant science*. 11:293. doi: 10.3389/fpls.2020.00293
- Hirayama, T. and Shinozaki, K. (2010). Research on plant abiotic stress responses in the post-genome era: past, present and future. *Plant Journal*, 61, 1041-1052. doi: 10.1111/j.1365-313X.2010.04124.x
- Hu, H. and Xiong, L. (2014). Genetic engineering and breeding of drought-resistant crops. *Annual review of plant biology*, 65, 715-741.
- Jung, J. K. H. M. and McCouch, S. R. M. (2013). Getting to the roots of it: genetic and hormonal control of root architecture. *Frontiers in plant science*, 4, 186.
- Kudla, J., Batistic, O. and Hashimoto, K. (2010). Calcium signals: the lead currency of plant information processing. *Plant Cell*, 22, 541-563. doi: 10.1105/tpc.109.072686

- Lanceras, J. C., Pantuwan, G., Jongdee, B. and Toojinda, T. (2004). Quantitative trait loci associated with drought tolerance at reproductive stage in rice. *Plant Physiology*, 135(1), 384-399.
- Lata, C. and Prasad, M. (2011). Role of DREBs in regulation of abiotic stress responses in plants. *Journal of Experimental Botany*, 62, 4731–4748. doi: 10.1093/jxb/err210
- Levitt, J. (1980). *Responses of plants to environmental stresses*. 2nd edn. (Academic Press: New York).
- Li, X. P., Tian, A. G., Luo, G. Z., Gong, Z. Z., Zhang, J. S. and Chen, S. Y. (2005). Soybean DRE-binding transcription factors that are responsive to abiotic stresses. *Theoretical and Applied Genetics*, 110(8), 1355-1362.
- Li, Y., Ruperao, P., Batley, J., Edwards, D., Khan, T., Colmer, T. D. and Sutton, T. (2018). Investigating drought tolerance in chickpea using genome-wide association mapping and genomic selection based on whole-genome resequencing data. *Frontiers in plant science*, 9, 190.
- Lipiec, J., Doussan, C., Nosalewicz, A. and Kondracka, K. (2013). Effect of drought and heat stresses on plant growth and yield: a review. *International Agrophysics*, 27(4), 463-477.
- Lou, Q., Chen, L., Mei, H., Xu, K., Wei, H. *et al.* (2017). Root transcriptomic analysis revealing the importance of energy metabolism to the development of deep roots in rice (*Oryza sativa* L.). *Frontiers in plant science*, 8, 1314.
- Lu, P. L., Chen, N. Z., An, R., Su, Z., Qi, B. S., Ren, F. *et al.* (2007). A novel drought-inducible gene, ATAF1, encodes a NAC family protein that negatively regulates the expression of stress-responsive genes in Arabidopsis. *Plant molecular biology*, 63(2), 289-305.
- Luo, L., Xia, H. and Lu, B. R. (2019). Crop breeding for drought resistance. *Frontiers in plant science*, 10, 314.
- Mahajan, S. and Tuteja, N. (2005). Cold, salinity and drought stresses: an overview. *Archives of biochemistry and biophysics*, 444(2), 139-158.
- Mare, C., Mazzucotelli, E., Crosatti, C., Francia, E. and Cattivelli, L. (2004). Hv-WRKY38: a new transcription factor involved in cold-and drought-response in barley. *Plant molecular biology*, 55(3), 399-416.
- Martinez, J.P., Silva, H., Ledent, J.F. and Pinto, M. (2007). Effect of drought stress on the osmotic adjustment, cell wall elasticity and cell volume of six cultivars of common beans (*Phaseolus vulgaris* L.), *European Journal of Agronomy*, 26, 30-38.
- Masle, J., Gilmore, S.R. and Farquhar, G.D. (2005). The ERECTA gene regulates plant transpiration efficiency in Arabidopsis. *Nature* 436, 866–870.
-

- Mittal S., Arora K., Rao A. R., Mallikarjuna M. G., Gupta H. S. and Nepolean T. (2017). Genomic selection for drought tolerance using genome-wide SNPs in maize. *Frontiers in plant science*. 8:550 10.3389/fpls.2017.00550
- Morgan, J. M. (2000). Increases in grain yield of wheat by breeding for an osmoregulation gene: relationship to water supply and evaporative demand. *Australian Journal of Agricultural Research*, 51(8), 971-978.
- Murugesan, P., Aswathy, G. M., Sunil Kumar, K., Masilamani, P., Kumar, V. and Ravi, V. (2017). Oil palm (*Elaeis guineensis*) genetic resources for abiotic stress tolerance: A review. *Indian Journal of Agricultural Sciences*, 171, 12-17.
- Nakashima, K. and Yamaguchi-Shinozaki, K. (2013). ABA signaling in stress-response and seed development. *Plant Cell Reports*, 32, 959–970. doi: 10.1007/s00299-013-1418-1.
- Naresh Kumar, S. (2002). Agro-meteorology and drought management in coconut. In: Training manual on drought stress and its management in plantation crops. Central Plantation Crop Research Institute, Kasaragod, India.
- Naresh Kumar, S., Rajagopal, V. and Karun, A. (2000). Leaflet anatomical adaptations in coconut cultivars for drought tolerance. pp 225-229. In: N Muralidharan and R Raj Kumar (eds). *Recent advances in plantation crop research*. Allied Pub. Ltd, New Delhi, India.
- Nemeskéri, E., Molnár, K., Víg, R., Nagy, J. and Dobos, A. (2015). Relationships between stomatal behaviour, spectral traits and water use and productivity of green peas (*Pisum sativum* L.) in dry seasons. *Actaphysiologiae plantarum*, 37(2), 34.
- Nodichao, L., Chopart, J. L., Rounsard, O., Vauclin, M., Aké, S. and Jourdan, C. (2011). Genotypic variability of oil palm root system distribution in the field. Consequences for water uptake. *Plant and soil*, 341(1), 505-520.
- Oosten, M.J.V., Costa, A., Punzo, P., Landi, S., Ruggiero, A., Batelli, G. and Grillo, S. (2016). Genetics of Drought Stress Tolerance in Crop Plants. In: Hossain et al. (eds.), *Drought Stress Tolerance in Plants*, Vol 2, Springer International Publishing Switzerland. Pp: 39-70 DOI 10.1007/978-3-319-32423-4_2
- Perera P.I.P., Hoche L.K., Weerakoon D.M.D., Yakandawala S.C. and Fernando J.L.V. (2010). Early inflorescence and floral development in *Cocos nucifera* L. (Arecaceae: Arecoideae). *South African Journal of Botany* 76: 482-492
- Pirasteh-Anosheh, H., Saed-Moucheshi, A., Pakniyat, H. and Pessarakli, M. (2016). Stomatal responses to drought stress. *Water stress and crop plants*, 24-40.
- Rabey, H. E., Al-Malki, A., and Abulnaja, K. (2016). Proteome analysis of date palm (*Phoenix dactylifera* L.) under severe drought and salt stress. *International Journal of Genomics*. 2016:7840759.

- Rajagopal, V., Bai, K. K. and Kumar, N. (2005). Breeding for drought tolerance in coconut: Status and potentials. *Coconut Genetic Resources*, 282.
- Rajagopal, V., kashuri Bai, K.V. and Naresh Kumar, S. (2000). Drought management in plantation crops. Pp 30-35. In: P.Rethinam, Khan H.H, Reddy V.M, Mandal P.K and Sudash K (eds). Plantation crops research and development in the new millennium. *Abstracts of PLACROSYM XV*, 10-13 December 2002, Mysore, India.
- Ramesh, S. V., Hebbar, K. B., Rajesh, M. K., Sukumar, P. A., Raj, K. G. and Bobby, A. (2020). Transcriptome Analysis of *Cocos nucifera* L. Seedlings Having Contrasting Water-Use Efficiency (WUE) under Water-Deficit Stress: Molecular Insights and Genetic Markers for Drought Tolerance. *1st International Electronic Conference on Plant Science* Vol. (1), p. 15.
- Ranganayakulu, G. S., Veeranagamallaiah, G. and Sudhakar, C. (2013). Effect of salt stress on osmolyte accumulation in two groundnut cultivars (*Arachis hypogaea* L.) with contrasting salt tolerance. *African Journal of plant science*, 7(12), 586-592.
- Reddy, A. S., Ali, G. S., Celesnik, H. and Day, I. S. (2011). Coping with stresses: roles of calcium and calcium/calmodulin-regulated gene expression. *Plant Cell*, 23, 2010–2032. doi: 10.1105/tpc.111.084988
- Reguera, M., Peleg, Z. and Blumwald, E. (2012). Targeting metabolic pathways for genetic engineering abiotic stress-tolerance in crops. *Biochimica et Biophysica Acta (BBA)-Gene Regulatory Mechanisms*, 1819(2), 186-194.
- Sacala, E. (2009). Role of silicon in plant resistance to water stress. *Journal of Elementology*, 14(3), 619-630.
- Safronov, O., Kreuzwieser, J., Haberer, G., Alyousif, M., Schulze, W. et al. (2017). Detecting early signs of heat and drought stress in *Phoenix dactylifera* (date palm). *PLoS One* 12: e0177883. doi: 10.1371/journal.pone.0177883
- Sallam, A., Alqudah, A. M., Dawood, M. F., Baenziger, P. S. and Börner, A. (2019). Drought stress tolerance in wheat and barley: advances in physiology, breeding and genetics research. *International journal of molecular sciences*, 20(13), 3137.
- Samarasinghe, C. R. K., Meegahakumbura, M. K., Dissanayaka, H. D. M. A. C. and Perera, L. (2018a). Performance of the second generation of Ambakelle special variety under varying rainfall and climatic conditions. *Proceedings of the 7th Annual Research symposium of the Young Scientist Forum of the National Science and Technology Commission (NASTEC)*, Sri Lanka; pp: 122-125
- Samarasinghe, C. R. K., Meegahakumbura, M. K., Dissanayaka, H. D. M. A. C., Kumarathunge, D. and Perera, L. (2018b). Variation in yield and yield components of different coconut cultivars in response to within year rainfall and temperature variation. *Scientia Horticulturae*, 238, 51-57.
-

- Rajagopal, V., Bai, K. K. and Kumar, N. (2005). Breeding for drought tolerance in coconut: Status and potentials. *Coconut Genetic Resources*, 282.
- Samarasinghe, C. R. K., Meegahakumbura, M. K., Kumarathunge, D. P., Dissanayaka, H. D. M. A. C., Weerasinghe, P. R. and Perera, L. (2021). Genotypic selection approach made successful advancement in developing drought tolerance in perennial tree crop coconut. *Scientia Horticulturae*, 287, 110220.
- Ranasinghe C.S. (2013). Impact of climate change on coconut and identification of adaptation measures with special emphasis on reproductive development. In: Gunasena H.P.M., Gunathilake H.A.J., Everard J.M.D.T., Ranasinghe C.S. and Nainanayake N.P.A.D. (Eds), *Proceedings of the International Conference on Climate Change Impacts and Adaptations for Food and Environment Security 'Sustaining Agriculture under Changing Climate'*, pp 101-114. Coconut Research Institute, Lunuwila, Sri Lanka; Ministry of Environment and Renewable Energy, Sri Lanka, World Agroforestry Centre, Regional Office, New Delhi, India.
- Schachtman, D. P. and Goodger, J. Q. (2008). Chemical root to shoot signaling under drought. *Trends Plant Science*, 13, 281–287. doi: 10.1016/j.tplants.2008.04.003
- Sharma, P., Jha, A. B., Dubey, R. S. and Pessarakli, M. (2012). Reactive oxygen species, oxidative damage, and antioxidative defense mechanism in plants under stressful conditions. *Journal of botany*, Vol. 2012, Article ID 217037, doi:10.1155/2012/217037
- Shen, Y. G., Zhang, W. K., He, S. J., Zhang, J. S., Liu, Q. and Chen, S (2003). An EREBP/AP2-type protein in *Triticum aestivum* was a DRE-binding transcription factor induced by cold, dehydration and ABA stress. *Theoretical and Applied Genetics*, 106(5), 923-930.
- Stocker, T. F., Qin, D., Plattner, G. K., Alexander, L. V., Allen, S. K. *et al.* (2013). Technical summary. In *Climate change 2013: the physical science basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change* (pp. 33-115). Cambridge University Press.
- Suresh, K., Nagamani, C., Ramachandrudu, K. and Mathur R.K. (2010). Gas-exchange characteristics, leaf water potential and chlorophyll *a* fluorescence in oil palm (*Elaeis guineensis* Jacq.) seedlings under water stress and recovery. *Photosynthetica*, 48:430-436
- Szabados, L. and Saviouré, A. (2010). Proline: a multifunctional amino acid. *Trends in Plant Science*, 15, 89–97. doi: 10.1016/j.tplants.2009.11.009
- Taiz, L. and Zeiger, E. (2002). *Plant physiology*. 3rd edn. (Sinauer Associates: Sunderland, MA, USA).

- Tiburcio, A. F., Altabella, T., Bitrián, M. and Alcázar, R. (2014). The roles of polyamines during the lifespan of plants: from development to stress. *Planta*, 240(1), 1-18.
- Tondelli, A., Francia, E., Barabaschi, D., Aprile, A., Skinner, J. S. *et al.* (2006). Mapping regulatory genes as candidates for cold and drought stress tolerance in barley. *Theoretical and Applied Genetics*, 112(3), 445-454.
- Turner, N. C. (1979). Drought resistance and adaptation to water deficits in crop plants. *Stress physiology in crop plants*, 343-372.
- Ubara, U. E., Agho, C. A., Aye, A. I., Yakubu, M., Eke, C. R. and Asemota, O. (2017). Identification of drought tolerant progenies in oil palm (*Elaeis guineensis* Jacq.). *International Journal of Advanced Research in Biological Sciences*, 4, 120-127.
- Van Oosten, M. J., Costa, A., Punzo, P., Landi, S., Ruggiero, A., Batelli, G. and Grillo, S. (2016). Genetics of drought stress tolerance in crop plants. *Drought Stress Tolerance in Plants*, 2, 39-70.
- Vandeleur, R. K., Sullivan, W., Athman, A., Jordans, C., Gilliam, M., Kaiser, B. N. and Tyerman, S. D. (2014). Rapid shoot-to-root signalling regulates root hydraulic conductance via aquaporins. *Plant, Cell & Environment*, 37(2), 520-538.
- Vanderauwera, S., Vandenbroucke, K., Inzé, A., Van De Cotte, B., Mühlenbock, P., De Rycke, R. and Van Breusegem, F. (2012). AtWRKY15 perturbation abolishes the mitochondrial stress response that steers osmotic stress tolerance in Arabidopsis. *Proceedings of the National Academy of Sciences*, 109(49), 20113-20118.
- Verma, V., Foulkes, M.J., Worland, A.J., Sylvester-Bradley, R., Caligari, P.D.S. and Snape, J.W. (2004). Mapping quantitative trait loci for flag leaf senescence as a yield determinant in winter wheat under optimal and drought-stressed environments. *Euphytica*, 135, 255-263.
- Voltas, J., Lopez-Corcoles, H. and Borrás, G. (2005). Use of bi-plot analysis and factorial regression for the investigation of superior genotypes in multi-environment trials. *European Journal of Agronomy*, 22(3), 309-324.
- Wang, L., Lee, M., Ye, B. and Yue, G. H. (2020). Genes, pathways and networks responding to drought stress in oil palm roots. *Scientific reports*, 10(1), 1-13.
- Waidyaratne K.P. and Perera S.A.C.N. (2020). 'Extreme weather events and crop yields: a case study with coconut', In: *Adapting to Climate Change: A Sri Lankan perspective*. Caldera, H.I.U. and Perera, S.A.C.N. (Eds.), Institute of Biology Sri Lanka, Colombo. pp.153-166
-

- Wen, X. and Moriguchi, T. (2015). Role of polyamines in stress response in horticultural crops. In: *Abiotic stress biology in horticultural plants*. Kanayama, Y. and Kochetov, A. (eds) Springer, Japan, pp: 35–45.
- Wickramaratne, M. R. T. (1987). Breeding coconuts for adaptation to drought. *Coconut Bulletin* (Sri Lanka).
- Xu, Z. and Zhou, G. (2008). Responses of leaf stomatal density to water status and its relationship with photosynthesis in a grass. *Journal of experimental botany*, 59(12), 3317–3325.
- Yaish, M.W. (2015). Proline accumulation is a general response to abiotic stress in the date palm tree (*Phoenix dactylifera* L.). *Genetics and Molecular Research* 14: 9943–9950
- Yadollahi, A., Arzani, K., Ebadi, A., Wirthensohn, M. and Karimi, S. (2011). The response of different almond genotypes to moderate and severe water stress in order to screen for drought tolerance. *Scientia Horticulturae*, 129(3), 403–413.
- Yang, H. M. and Wang, G. X. (2001). Leaf stomatal densities and distribution in *Triticum aestivum* under drought and CO₂ enrichment. *Chinese Journal of Plant Ecology*, 25(3), 312.
- Zandalinas, S. I., Sales, C., Beltrán, J., Gómez-Cadenas, A. and Arbona, V. (2017). Activation of secondary metabolism in citrus plants is associated to sensitivity to combined drought and high temperatures. *Frontiers in Plant Science*. 7,1954. doi: 10.3389/fpls.2016.01954
- Zhao, C., Liu, B., Piao, S., Wang, X., Lobell, D. B. et al. (2017). Temperature increase reduces global yields of major crops in four independent estimates. *Proceedings of the National Academy of Sciences*. U.S.A. 114, 9326–9331. doi: 10.1073/pnas.1701762114
- Zheng, B. S., Yang, L., Zhang, W. P., Mao, C. Z., Wu, Y. R. et al. (2003). Mapping QTLs and candidate genes for rice root traits under different water-supply conditions and comparative analysis across three populations. *Theoretical and Applied Genetics*, 107(8), 1505–1515.
- Zhang, Y.P., Wang, Z.M., Wu, Y.C., and Zhang, X. (2006). Stomatal characteristics of different green organs in wheat under different irrigation regimes, *Acta Agronomica Sinica*, 32, 70–75.
- Zlatev, Z. and Lidon, F. C. (2012). An overview on drought induced changes in plant growth, water relations and photosynthesis. *Emirates Journal of Food and Agriculture*, 24, 57–72. doi: 10.9755/ejfa.v24i1.10599

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A. B. Perera^{1,2} and M. N. Silva^{3*}

A. B. Perera^{1,2}, M. N. Silva^{3*} and R. T. Alahakoon⁴

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- Page 2: Text

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$$a = b + c + d + e + f + g + h + i + j + k + l + m + n + o + p + q + r + s + t + u + v + w + x + y + z \quad (2)$$

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12. If the quoted passage is in the middle of a sentence, end the passage with quotation marks, cite the source in parentheses immediately, and continue the sentence.

Example: Fifty percent “of spontaneous speech is estimated to be non-speech” (Shaughnessy, 2020, p. 24).

13. If the quotation is longer than forty words, it is set off without quotations marks in an indented block (double spaced). The source is cited in parentheses after the final period.

Example: This is further explained by Shaughnessy's (2020) following statements:

In a speech, pauses mark rates of respiration, set off certain words for rhetorical emphasis, facilitate phonological maneuvers, regulate the rhythms of thought and articulation and suggest grammatical structure. Modern punctuation, however, does not provide a score for such a complex orchestration. (p. 24)

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